



Reducing dietary crude protein levels while meeting metabolizable protein requirements: Performance of dairy cows over a full lactation period

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ABSTRACT

A total of 90 Holstein dairy cows (24 primiparous, 66 multiparous, mean parity = 3.0) were fed diets containing either 150, 160, or 170 g CP/kg DM from 8 to 180 DIM, with all diets designed to supply at least 100% MP requirements. On d 181, half of the cows in each treatment changed to a diet containing 140 g CP/kg DM (supplying 100% MP requirements), with the remaining cows continuing to be offered their original treatment diets. This resulted in 6 treatments in the mid-late lactation period (181–280 DIM): 150, 150/140, 160, 160/140, 170, and 170/140 g CP/kg DM. Decreasing dietary CP concentration from 170 to 150 g CP/kg DM did not affect DMI, milk yield, milk fat or protein yield in early lactation (8–181 DIM), but reduced MUN and the ratio of n-6 to n-3 fatty acids in milk, as well as serum albumin, globulin, total protein, and urea concentrations. In addition, reducing dietary CP content from 170 to 160 g CP/kg DM improved N use efficiency (NUE; milk N/N intake), with no further improvement with 150 g CP/kg DM. Treatment had no effect on apparent total-tract ration digestibility in early lactation. Urinary N output decreased with decreasing dietary CP content in early lactation, whereas manure N (fecal N plus urinary N) output increased. Urinary N/manure N decreased from 170 to 160 g CP/kg DM diet, whereas no further reduction was observed with the 150 g CP/kg DM diet. Cows that remained on the 150 g CP/kg DM treatment in mid-late lactation (181–280 DIM) had a lower DMI than those which remained on the diet containing 170 g CP/kg DM. Reducing the dietary CP concentration to 140 g CP/kg DM in mid-late lactation reduced DMI, milk yield, and milk fat and protein yields, compared with offering the 170 or 160 g CP/kg DM diet throughout lactation, possibly reflecting a response to oversupply of MP with the latter treatments, rather than an undersupply of MP with the former, although this is

uncertain. Concentrations of C18:2 *cis*-9,*trans*-11 and the ratio of n-6 to n-3 fatty acids in milk were lower for cows offered diets containing 140 or 150 g CP/kg DM in mid-late lactation compared with 160 or 170 g CP/kg DM. Reducing dietary CP concentration from 170 to 140 g CP/kg DM improved NUE from 0.28 to 0.34 in mid-late lactation. Nitrogen digestibility was reduced when cows were offered the 140 g CP/kg DM diet compared with the 150 or 160 g CP/kg DM diet. Lowering the dietary CP concentration from 170 to 140 g CP/kg DM in mid-late lactation decreased N output in milk, feces, urine, and manure. These results indicate that a dietary CP content of 160 g CP/kg DM, which met the MP requirement of cows, may be optimal to support performance over a whole lactation, improve NUE, and reduce N excretion, thus contributing to a more sustainable approach to dairy cow production.

Key words: dairy cow, protein, nitrogen, nitrogen use efficiency, lactation

INTRODUCTION

The nitrogen use efficiency (NUE; defined as nitrogen N output in milk/N consumed) of dairy cows within grassland-based systems has been estimated to be less than 25% (Chen et al., 2020). Excreted N, in particular urinary N, is at risk of being lost to the atmosphere as ammonia and nitrous oxide and to waterways as nitrates (Hristov et al., 2011; Bougouin et al., 2016; Wattiaux et al., 2019). Given the strong positive relationships between dietary N content and intake and urinary N excretion (Barros et al., 2017), lowering the CP content of the diet can improve NUE and reduce N loss to the environment. However, although reducing the CP content of dairy cow diets can lower feed costs, wide-scale adoption will only be acceptable if accompanied by minimal or no loss in productivity.

The impact of reducing the CP content of dairy cow diets has been examined in many short-term studies. For example, in a change-over design study with early- to mid-lactation cows, Olmos Colmenero and Broderick

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-24. Nonstandard abbreviations are available in the Notes.

(2006) reported no difference in DMI or milk yield, but a reduction in N excretion, when dietary CP content was reduced from 194 to 135 g CP/kg DM. However, Katongole and Yan (2020) found that reducing dietary CP from 177 to 141 g CP/kg DM decreased milk yield and N excretion and improved NUE. Furthermore, in an 84-d study involving late-lactation dairy cows, Barros et al. (2017) found that reducing dietary CP content from 162 to 144 g CP/kg DM had no effect on milk yield, whereas a further reduction in CP to 131 or 118 g/kg DM decreased milk yield. However, the short-term nature of these studies means there is little time for body pools of N to adapt to dietary changes, and although concerns about this have been raised by several authors (Lapierre and Lobley, 2001; Firkins and Reynolds, 2005), there is no unanimity about the importance of conducting long-term studies (Liu et al., 2021). A small number of studies have examined the impact of dietary CP content over at least 1 full lactation. For example, Reynolds et al. (2018) offered diets containing 140, 160, and 180 g CP/kg DM over 3 full lactations, with diets designed to either undersupply, meet, or oversupply MP requirements. Milk production and fertility were lower with the diet containing 140 g CP/kg DM, whereas milk production did not differ between the diets containing 160 and 180 g CP/kg DM, although NUE was greater with the former. Similarly, in a full-lactation study involving 3 dietary CP levels (114, 144, and 173 g CP/kg DM) milk yield, milk fat, milk protein and milk fat plus protein yields showed a linear reduction with decreasing protein levels (Law et al., 2009). However, this study also provided evidence that dietary CP levels could be reduced further in later lactation with no loss in performance. For example, from lactation d 151 to 305, a subgroup of cows from the 173 g CP/kg DM treatment had their diet CP level reduced to 144 g CP/kg DM with no loss in performance. Similarly, Wu and Satter (2000) observed that neither milk yield nor 3.5% fat corrected milk yield differed between cows offered a diet with a CP content of 174 g CP/kg DM in the first 16 wk of lactation and 160 g CP/kg DM from wk 17 to 44 of lactation, compared with cows offered diets containing 179 g CP/kg DM during late lactation.

Although Law et al. (2009) did not present MP supply or requirements, a retrospective assessment of these diets suggests that all may have undersupplied MP in early lactation. Thus, the current study was designed to examine performance and nutrient utilization when cows were offered diets containing a range of CP concentrations (150, 160, and 170 g CP/kg DM), but with all diets designed to meet the cows' MP requirements in early lactation. It was hypothesized that NUE in early lactation could be improved by reducing diet CP levels, without loss in performance, provided that MP requirements were met. In addition, the study sought to re-examine the findings of

Law et al. (2009) that dietary CP levels could be reduced in mid-late lactation with no loss in performance.

MATERIALS AND METHODS

This study was conducted at the Agri-Food and Biosciences Institute (AFBI; Hillsborough, Northern Ireland; 54°27'N, 06°04'W). Ethical approval for the work was given by the Animal Welfare Ethical Review Body at AFBI Hillsborough, with all experimental procedures conducted under an experimental license (PPL2877) granted by the Department of Health, Social Services and Public Safety for Northern Ireland in accordance with the Animals (Scientific Procedures) Act 1986.

Experimental Animals and Housing

This study involved 90 Holstein dairy cows (24 primiparous and 66 multiparous, mean parity 3.0, SD $\pm$ 1.15) which calved between October 19 and December 28, 2020. Cows were randomly allocated to 1 of 3 treatments before calving, with a check made to ensure that treatment groups remained balanced for calving date (<7 d); parity (primiparous and multiparous cows allocated separately); PTA for milk yield, fat yield (SD $\pm$ 0.4 kg), and protein yield (SD $\pm$ 0.3 kg); fat percentage and protein percentage (SD $\pm$ 0.02%); previous lactation (305-d) milk yield (SD $\pm$ 30 kg), milk fat content (SD $\pm$ 0.1%), and protein content (SD $\pm$ 0.05%; multiparous cows only); and BCS (SD $\pm$ 0.1) and BW (SD $\pm$ 10 kg) measured during the 2-wk period before calving. Cows had a mean PTA₂₀₂₀ (PTA for the year 2000) for milk yield, fat yield, and protein yield of 181, 11.8, and 10.5 kg, respectively.

For the duration of the 8-wk dry period, cows were offered a grass silage (mean CP of 119 g/CP kg DM [SD $\pm$ 1.63] and ME of 10.2 MJ/kg [SD $\pm$ 0.62]) ad libitum, with a dry-cow mineral and vitamin mix (crude ash 96%; calcium 1.66%; phosphorus 3%; magnesium 25%; sodium 14.5%; 400,000 IU/kg vitamin A; 100,000 IU/kg vitamin D₃; 8,000 IU/kg vitamin E; 3,000 mg/kg Cu; 50 mg/kg Se; 4,500 mg/kg Zn; 2,000 mg/kg Mn; 80 mg/kg Co; 500 mg/kg I; Trouw Nutrition, Cheshire, UK) mixed with the silage to achieve a target intake of 125 g/cow per day. During the last 3 wk before calving (close-up period), calcined magnesite (Trouw Nutrition, Cheshire, UK) was added to the mix to achieve a target intake of 75 g/cow per day. Cows were moved to a straw-bedded pen ~24 to 48 h before calving based on behavioral observations. Following calving (normally within 24 h), cows were transferred to an experimental freestall house where they had access to individual cubicles that were fitted with rubber mats and bedded with sawdust. The cubicle to cow ratio was $\geq$ 1.1 at all times. The house had

a solid concrete floor which was cleaned every 3 h using an automated scraper system. Cows had access to fresh water at all times.

Treatments

The study was conducted from calving until 280 DIM. During the first 7 d following calving, all cows were offered a common diet containing 160 g CP/kg DM to allow collection of baseline rumen fluid samples (A. Zeleke, Queen's University Belfast, Belfast, UK, unpublished data). Thereafter (8–180 DIM), cows were offered 1 of 3 dietary treatments which were designed to contain either 150, 160, or 170 g CP/kg DM. All diets were formulated to supply at least 100% of cows estimated MP requirements based on equations within Feed into Milk (FiM; Thomas, 2004), the current feed rationing system for dairy cows within the United Kingdom. Rations offered in early lactation (8–179 DIM) were formulated for multiparous cows based on the following assumptions across all treatments: 40 L of milk, 4.25% milk fat, 3.3% milk protein, 630 kg BW, and 24.6 kg DMI, with the MP requirement estimated to be 2,683 g/d. Diets were formulated to provide ~9.4% RDP, 328 g NDF/kg DM, 237 g ADF/kg DM, 153 g starch/kg DM, and 12.3 MJ ME/kg DM). At 180 DIM, cows within each treatment were divided into 2 balanced groups (balanced for calving date and parity, as well as for milk yield, milk fat content, milk protein content, BCS, and BW during the previous 7 d), with half of the cows from each treatment remaining on their original diet and the other half changing to a diet containing 140 g CP/kg DM. Thus, 6 treatments were examined during 181 to 280 DIM: 150, 150/140, 160, 160/140, 170, and 170/140 g CP/kg DM. The diet containing 140 g CP/kg DM was designed to meet 100% of the cows' estimated MP requirements in mid-late lactation. Diets were formulated for multiparous cows based on the following assumptions across all treatments: 33.0 L of milk, 4.4% milk fat, 3.5% milk protein, 670 kg BW, and 23.8 kg DMI, with MP requirement estimated to be 2,337 g/d. The 140 g CP/kg DM diet was formulated to provide ~9.0% RDP, 337 g NDF/kg DM, 238 g ADF/kg DM, 182 g starch/kg DM, and 12.2 MJ ME/kg DM).

Throughout the experiment, all cows were offered a partial mixed ration comprising equal proportions of grass silage and concentrates (50:50 on a DM basis), and chopped straw (treated using a Roto Grind [Enegis Ltd., Warwickshire, UK]), which was included in the ration to achieve a nominal target intake of 0.32 kg DM/cow per day. The grass silage offered from the time the study commenced until February 4, 2021, was prepared from primary growth herbage, whereas grass silage offered after February 4, 2021, was prepared from a mix of primary growth and primary regrowth herbage (so as to maintain

forage protein content at the target level). The concentrates offered with each treatment differed in ingredient and chemical composition to achieve the required total diet CP levels. Concentrates were formulated and total rations balanced according to the equations within FiM using NutriOpt software (Nutreco, Trouw Nutrition, Amersfoort, the Netherlands), which in seeking to meet MP requirements, takes account of fermentable energy, total fermentable carbohydrates, and protein balance. Variation in silage composition within and between silos necessitated that the ingredient composition of the concentrates offered be adjusted on several occasions during the study so as to maintain total diet CP levels. The mean ingredient composition of each experimental concentrate, weighted for the mean number of days that each concentrate formulation was offered for, is presented in Table 1.

The rations were prepared daily at ~0900 to 1000 h and offered *ad libitum* at 107% of the previous day's intake. Uneaten ration was removed the following day at ~0800 to 0900 h. Rations were prepared using a paddle-type diet feeder wagon (Redrock, Armagh, Northern Ireland). The total quantity of silage and straw required for all treatments was initially mixed for ~5 min and then deposited on a clean silo floor. The quantity of this mix required for each individual treatment was then removed from this

Table 1. Mean ingredient composition of concentrates offered as part of diets designed to supply 140, 150, 160, and 170 g CP/kg of total diet DM (g/100 g, fresh basis)¹

Item	Total diet CP content ² (g/kg DM)			
	140	150	160	170
Maize meal (ground)	34.6	30.8	26.9	23.1
Wheat (rolled)	11.8	10.5	8.5	6.6
Barley (rolled)	6.6	6.7	6.9	7.0
Soybean meal (high protein)	8.0	10.1	9.7	9.2
Rapeseed meal	5.8	8.7	11.3	13.9
Soy hulls	12.9	11.2	9.4	7.5
Sugar beet pulp	12.5	12.5	12.5	12.5
Maize distillers	—	2.0	5.0	8.0
Protected fat (Megalac ³)	2.3	2.3	2.3	2.3
Sugarcane molasses (liquid)	2.0	2.0	2.0	2.0
Dicalcium phosphate	0.6	0.4	0.2	—
Protected soy protein (Sopralin ⁴)	—	—	2.5	5.0
Mineral-vitamin premix ⁵	3.0	3.0	3.0	3.0

¹Values presented represent the mean ingredient composition of each concentrate offered (weighted for the mean number of feeding days each concentrate was offered for). Formulations were modified during the study to ensure total diet CP levels were maintained.

²Crude protein content refers to the total dietary CP of the experimental diet (forage + concentrate).

³Volac Wilmar Feed Ingredients Ltd., Hertfordshire, UK.

⁴Trouw Nutrition Ireland, Belfast, Northern Ireland, UK.

⁵Premix provided (per kg) vitamin A, 10,000 IU; vitamin D₃, 2,000 IU; vitamin E, 90 IU; Se, 0.55 mg; Cu, 32.5 mg; Co, 0.65 mg; I, 7.0 mg; Mn, 80.0 mg; Zn, 96 mg; Trouw Nutrition Ireland, Belfast, Northern Ireland, UK.

pile and placed back in the diet feeder, the required quantity of the appropriate concentrate was added to the mix, and the forage and concentrate were mixed for 5 min. The rations were then transferred from the diet feeder to a series of feed boxes mounted on weight scales (Controlling and Recording Feed Intake, BioControl, Rakkestad, Norway), with cows accessing food in these boxes via an electronic identification system, enabling individual cow intakes to be recorded daily. The order in which the diets were prepared was rotated on a daily basis, and the feed boxes each treatment group had access to were rotated every 3 wk to minimize any confounding impact of feeding time and housing environment on performance. In addition, throughout the experiment, all cows were offered a common commercial concentrate pellet during milking (CP, 182 g/kg DM; NDF, 263 g/kg DM; ADF, 128 g/kg DM; starch 291 g/kg DM; ME, 13.2 MJ/kg DM), 0.25 kg at each milking, with an ingredient composition as follows (g/100 g fresh basis): wheat, 17.4; maize meal, 17.5; maize distillers, 8.5; maize gluten, 11.0; sugar beet pulp, 6.1; soybean meal (high protein), 8.6; soy hulls, 17.45; sugarcane molasses (liquid), 8.0; palm fatty acid distillate, 0.95; mineral and vitamin premix, 4.5 (vitamin A, 8,000 IU/kg; vitamin D₃, 3,000 IU/kg; vitamin E, 50 IU/kg; Se, 0.55 mg/kg; Cu, 25.0 mg/kg; Co, 0.47 mg/kg; I, 5.0 mg/kg; Mn, 60 mg/kg; Zn, 100 mg/kg).

Cow Measurements

Feed intakes were measured as described above. All cows were milked twice daily (between 0600 and 0800 h and between 1500 and 1700 h) throughout the experiment using a 50-point rotary milking parlor (Boumatic, Madison, WI). Milk yields were automatically recorded at each milking, and a total daily milk yield for each cow for each 24-h period was calculated. Milk samples were collected during 2 consecutive milkings (p.m. and a.m.) each week, treated with a preservative tablet (Broad Spectrum Microtabs II, Advanced Instruments, Norwood, MA), and stored at 4°C until analyzed. Milk samples were analyzed for fat, protein, lactose, and urea nitrogen concentrations using a mid-infrared milk analyzer (Milkoscan Combifoss 7, Foss Electric, Hillerød, Denmark), and a weighted concentration of each constituent determined for each 24-h sampling period.

In addition, a further milk sample was collected from each cow at 60 (± 10 d), 160 (± 14 d) and 240 (± 18 d) DIM, during 2 consecutive milkings and bulked to provide a representative daily sample. The sample was then stored at -20°C until analyzed for milk fatty acids (FA), as follows: milk fat was extracted from 1.0 mL of homogenized milk using a chloroform methanol extraction method (Bligh and Dyer, 1959), and FA determined as methyl esters (FAME). The FA composition was deter-

mined using GLC, with an aliquot (1.0 μL) of the FAME extract injected onto a CP Sil88 capillary column (100 m $\times$ 0.25 mm i.d. $\times$ 0.2 μm film thickness) in an Agilent 7890 gas chromatograph (both Agilent Technologies, Santa Clara, CA), equipped with a temperature programmable injector operated in the split mode and a flame ionization detector. The oven was initially held at 50°C for 4 min, then increased at $8^{\circ}\text{C}/\text{min}$ to 110°C , then $5^{\circ}\text{C}/\text{min}$ to 170°C (hold time 10 min), and finally increased at $2^{\circ}\text{C}/\text{min}$ to 225°C (hold time 30 min). Fatty acids were identified by their retention time with reference to commercially available FA standards (37 Supelco FAME mix) and individual standards for those not in the mix (Sigma-Aldrich Co. Ltd., Gillingham, UK), and were quantified using C13 FAME as an internal standard.

Cow BW was recorded twice daily from calving until the end of the study (immediately after each milking) using an automated weighbridge, with a mean weekly BW determined for each cow. Throughout the experiment, the BCS of each cow was assessed fortnightly by a trained technician as described by Edmonson et al. (1989).

Estimated energy balance (MJ/d) of each cow was calculated weekly using equations contained within FiM dairy cow rationing system (Agnew et al., 2004), where mean energy balance was determined using the following equation:

mean energy balance =

$$\left\{ \left[ME_{\text{main+milk}} \times BW^{0.75} \right] + \left[\frac{(0.0013 \times BW)}{K_m} \right] + ME_c - 10 \right\} - ME_i$$

where $ME_{\text{main+milk}}$ is the ME required for maintenance and milk production (MJ/kg metabolic weight), $BW^{0.75}$ is metabolic BW, K_m is the efficiency of utilization of ME for activity [calculated as $0.35 \times \text{ME (MJ)/gross energy (MJ)} + 0.503$], ME_c is the ME required for pregnancy (MJ/cow per d), and ME_i is ME intake (MJ/cow per d). Data for mean daily milk yield, milk fat, protein and lactose concentrations, and mean BW were used in the calculations for the energy balance variables (ME requirements for maintenance, milk production, pregnancy, activity, and ME intake).

Blood samples were collected from the tail of each cow between 0900 and 1100 h (before cows had access to fresh food) at wk 2, 4, 6, 8, 10, 14, 18, 28, 32, and 38 of lactation. Two samples were collected, 1 in a heparin-coated tube and 1 in a fluoride oxalate tube, and centrifuged ($1,811 \times g$ for 15 min at 5°C) to isolate plasma. Plasma samples were stored at -20°C until analysis.

Plasma BHB concentrations were determined using the oxidation of D-3-hydroxybutyrate to acetoacetate by the enzyme D-3-hydroxybutyrate dehydrogenase (GlenBio Ltd., Antrim, UK), and plasma glucose concentrations

were determined using a colorimetric, enzymatic method with glucose oxidase (Cormay Diagnostics, Warsaw, Poland). Plasma nonesterified fatty acid (NEFA) concentrations were determined by the conversion of NEFA in a sample to acyl-CoA, AMP, and pyrophosphoric acid by the action of acyl-CoA synthetase, under coexistence with coenzyme A and adenosine 5-triphosphate disodium salt (GlenBio Ltd., Antrim, UK). Plasma urea concentrations were analyzed using the kinetic enzymatic method with urease and glutamate dehydrogenase (Cormay Diagnostics, Warsaw, Poland). Total protein concentrations were analyzed using the biuret reaction (Cormay Diagnostics, Warsaw, Poland) and plasma albumin concentrations were determined via the bromocresol green method (Cormay Diagnostics, Warsaw, Poland). Plasma globulin concentrations were calculated by subtracting the concentration of albumin from total protein concentration. Plasma creatinine concentrations were analyzed using a modified version of the Jaffe reactions without deproteinization (Cormay Diagnostics, Warsaw, Poland). All blood metabolites were analyzed on the Biolis 24i Premium clinical chemistry analyzer (Tokyo Boeki Medisys Inc., Hino, Japan).

Backfat thickness and gluteus medius (GM) muscle thickness at the sacral area adjacent to the first coccygeal vertebrae, and longissimus dorsi muscle thickness at the position of the fifth lumbar vertebrae and between the 12th and 13th ribs, were measured ultrasonographically as described by Megahed et al. (2019). Measurements were recorded weekly for each cow for the first 10 wk postpartum, and once every 4 wk thereafter for the duration of the experiment. Hair was clipped at the measurement sites, and vegetable oil applied to facilitate transducer contact. A portable ultrasonographic unit (Easiscan, IMV imaging, Scotland, UK), with a 4.0 to 8.5 MHz linear transducer, using bovine meat quality measurement software available, was applied. Suitable ultrasonographic images were obtained, and the corresponding backfat and muscle depth measurements generated on the ultrasonographic unit were recorded.

Feed Analysis

The grass silage offered was sampled daily throughout the experiment and dried at 60°C for 48 h to determine oven DM (ODM) content. Samples of dried silage were collected thrice weekly and bulked for each 14-d period, with the bulked sample milled through a sieve with a 0.8-mm aperture and analyzed for NDF, ADF, and ash concentrations. In addition, a fresh silage sample was analyzed weekly for gross energy (GE), N, pH, ammonia-N, and volatile components, and ME concentration was predicted using near infrared reflectance spectroscopy according to Park et al. (1998). The straw offered

was sampled weekly, with samples bulked for each 28-d period. Each bulked sample was dried at 60°C for 48 h to determine oven DM content, with the dried sample milled through a sieve with a 0.8-mm aperture and analyzed for concentrations of N, NDF, ADF, ash, and GE. Each concentrate offered was sampled weekly and dried at 60°C for 48 h to determine oven DM content. Weekly samples of dried concentrate were subsequently milled through a 0.8-mm sieve, and analyzed for N, NDF, ADF, ash, and GE, whereas samples analyzed for starch concentrations were milled further through a sieve with a 0.5-mm aperture. All chemical analysis of feedstuffs offered were undertaken as described by Lavery et al. (2023). A bulked sample of each feedstuff (silage ($n = 7$), concentrate blends ($n = 16$ for each of the 150, 160, and 170 g CP/kg DM concentrates and $n = 7$ for the 140 g CP/kg DM concentrate), commercial concentrate ($n = 7$), and straw ($n = 4$) were analyzed for AA profiles. Samples were prepared by combining the dried milled samples collected weekly (concentrate blend), fortnightly (grass silage and concentrate pellets), or monthly (straw) in proportion to the length of time they were offered to cows. Amino acid concentrations were determined using the ion-chromatography-UV method in a commercial laboratory (Public Analyst Scientific Services Ltd., Wolverhampton, UK). In summary, the methodology involved hydrolysis of samples in aqueous hydrochloric acid, pH was adjusted, and the sample brought to volume with loading buffer and filtered. Amino acids were separated in an AA analyzer and the detection carried out using post-column derivatization with ninhydrin reagent and 440 and 570 nm. For the analysis of Cys and Met, samples were oxidized with hydrogen peroxide and formic acid to prevent loss during the acid hydrolysis step, and subsequent analysis was carried out as described previously.

Nutrient Utilization

During early lactation, 6 cows (2 primiparous and 4 multiparous: mean 92 DIM) from each of the 3 CP treatments (150, 160, and 170 g CP/kg DM), were used in a nutrient utilization study. Similarly, during mid-late lactation, 5 cows (2 primiparous and 3 multiparous: mean 240 DIM) from each of the 4 CP treatments (140, 150, 160, and 170 g CP/kg DM), were used in a nutrient utilization study. The 5 cows from the 140 g CP/kg DM treatment were selected from across those which had been managed on the 3 CP treatment groups in early lactation (2 cows from each of 150 and 160 g CP/kg DM treatments, and 1 cow from 170 g CP/kg DM treatment). Each treatment group was selected to be representative of the mean of the whole treatment group at that time, based on daily milk yield, milk composition, BW, and BCS. Cows were transferred from the main experimental cow house to a

metabolism unit for apparent total-tract diet digestibility and nutrient utilization measurements. Cows were tied by the neck in individual stalls fitted with a rubber mat and continued to be offered their experimental rations via a feedbox located at the front of each stall. Rations, as described for the main experiment, were prepared individually for each cow using a mini-mixer and offered daily (at ~107% of the previous day's intake) at 0900 h. Uneaten ration was removed the following day at 0800 h. As per the main experiment, 0.25 kg of the commercial concentrate (parlor pellet) was offered at each milking via a plastic feed bucket, which was placed within the feedboxes during milking (0700 and 1600 h). Cows had free access to fresh water at all times via a drinker located within each stall and were milked twice daily starting at 0700 and 1600 h.

Measurements of nutrient utilization commenced 24 h after cows were placed in the digestibility unit, and comprised a 6-d feeding period, followed 48 h later by a 6-d total feces and urine collection period. Feces were collected in plastic collection trays (96 cm × 108 cm × 36 cm deep) placed behind each cow. Urine was collected into a 25-L plastic container via a flexible plastic tube which was attached to a urine separation system and held in position over the vulva using hook-and-loop fasteners on a patch glued (Bostik, Colombes, France) to either side of the cows' tail head. On d 1, 2, 4, and 5, 300 mL of 50% sulfuric acid was added to the urine collection containers to minimize N losses as ammonia. On d 3 and 6, the containers were placed within insulated boxes full of ice to minimize N losses as ammonia, but avoiding the use of acid. This alternative methodology was necessary because feces and urine collected on these 2 d were used in a subsequent study designed to examine the impact of diet on gaseous fluxes from manure (X. Chen, Agri-Food and Biosciences Institute, Hillsborough, UK, unpublished data). These measurements would not have been possible if the urine had been preserved using acid. The total weight of feces and urine produced during each 24 h collection period was recorded, and a sample of each (5% of weight) taken and stored in a fridge at 4°C. Fecal and urine samples collected from d 1 to 3 and d 4 to 6 were subsequently bulked for analysis, with fresh feces and urine analyzed for N content, and freeze-dried urine samples analyzed for GE content. Feces samples were dried at 60°C for 96 h to determine ODM content, with the dried samples subsequently milled using a 0.8 mm sieve and analyzed for ADF, NDF, ash, and GE concentrations. Milk samples were taken at each milking (5% of yield), with samples from 2 consecutive milkings (p.m. and a.m.) bulked to provide a daily sample which was analyzed for N content. Milk samples were bulked for d 1 to 3 and d 4 to 6 and freeze-dried samples analyzed for GE concentrations. Concentrations of NDF and ADF

in feces were determined using a Fibertec analyzer (Fibertec FT122, Foss, Hillerød, Denmark) based on the method of Van Soest (1976), and ash concentrations were determined following combustion in a muffle furnace at 550°C for ~10 h. Nitrogen concentration of urine, milk, and fresh feces were determined using the Kjeldahl method (Tecator Kjeltac Auto 2400/2460 analyzer/sampler system, Foss). Gross energy concentrations of feces, freeze-dried milk, and urine samples were determined using a bomb calorimeter (Parr 6400 Bomb Calorimeter).

Silage offered during each digestibility study was sampled daily and analyzed for oven DM, GE, N, and volatile components. Dried silage samples were bulked for d 1 to 3 and d 4 to 6 of each study and subsequently analyzed for ADF, NDF, and ash concentrations. Straw and concentrates offered were sampled daily and analyzed for oven DM, with dried samples from each digestibility study bulked and subsequently analyzed for GE, NDF, ADF, N, and ash concentrations. A sample of the ration refused by each cow was taken daily and analyzed for oven DM content. The chemical analyses of all feedstuffs offered was as described for the main feeding study.

Statistical Analysis

Four cows were removed during the first 12 wk of the study for lameness ($n = 1$), intestinal ulcer ($n = 1$), and injury ($n = 2$) and were treated as missing values in the statistical analysis. A further 4 cows were removed from the study (lameness, $n = 2$; injury, $n = 1$; and mouth abscess, $n = 1$) at wk 23, 23, 34, and 37, with their data until the time of removal included in the statistical analysis, and with data after removal treated as missing values in the statistical analysis. Data were analyzed using Genstat v21 (VSN International Ltd.). Weekly milk production and milk composition data (8–180 DIM) were analyzed using a mixed model repeated measures (autoregressive model of order 1) analysis (REML estimation method), with week (wk 2–26) included as the repeated measure and cow as the subject factor; treatment (150, 160, and 170 g CP/kg DM) as a fixed effect; and lactation number (1, 2, 3, 4+) as an additional fixed effect, as

$$y_{ij} = m + lact_i + value305_i + trt_t + week_j + trt \times week_{ij} + animal_i \times week_j,$$

where y_{ij} is the response to the variable y for the animal i in week j , m is the overall mean, $lact_i$ is the lactation effect for animal i , $value305_i$ is the deviation from the appropriate previous lactation d 305 value for animal i , trt_t is the main effect of treatment t on animal i , $week_j$ is the main effect of week j , $trt \times week_{ij}$ is the interaction effect

of treatment t on animal i in week j , and $animal_i \times week_j$ is the residual effect of animal i in week j . Lactation number ($lact_i$) and deviation from the appropriate previous lactation d 305 value ($value305_i$, 0 used for primiparous cows) were used as covariates for all variables.

Weekly BW and fortnightly BCS data were analyzed using a similar model, but with the data from wk 1 used as a covariate in the analysis, with the following model:

$$y_{ij} = m + lact_i + y_wk1_i + trt_t + week_j + trt \times week_{ij} + animal_i \times week_j,$$

where y_{ij} is the response to the variable y for the animal i in week j , m is the overall mean, $lact_i$ is the lactation effect for animal i , y_wk1_i is the effect of variable y in wk 1 on subsequent values of y , trt_t is the main effect of treatment t on animal i , $week_j$ is the main effect of week j , $trt \times week_{ij}$ is the interaction effect of treatment t on animal i in week j , and $animal_i \times week_j$ is the residual effect of animal i in week j .

Weekly DMI (8–180 DIM), NUE, calculated apparent energy balance, blood metabolite concentrations (wk 2, 4, 6, 8, 10, 14, and 18 of lactation), and backfat and muscle depth measurements (wk 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 14, 18, 22, and 26) were analyzed using a similar model, but without the use of previous lactation data as a covariate. The correlation between weeks was modeled using an autoregressive model of order 1 if the time points were evenly spaced or an antedependence model of order 1 otherwise.

For mid-late lactation data (181–280 DIM), the response variables (weekly DMI, milk yield and milk composition, BW, and calculated apparent energy balance, NUE data, fortnightly BCS data, blood metabolites [wk 28, 32, and 38], and backfat thickness/muscle thickness [wk 30, 34, and 38]) were modeled using linear mixed methodology in a repeated measures design, with week of study (wk 27–40) + treatment (150, 150/140, 160, 160/140, 170, 170/140 g CP/kg DM) + week of study $\times$ treatment as fixed effects, with lactation number (1, 2, 3, 4+) fitted as additional fixed effect as

$$y_{ij} = m + lact_i + trt_t + week_j + trt \times week_{ij} + animal_i \times week_j,$$

where y_{ij} is the response to the variable y for the animal i in week j , m is the overall mean, $lact_i$ is the lactation effect for animal i , trt_t is the main effect of treatment t on animal i , $week_j$ is the main effect of week j , $trt \times week_{ij}$ is the interaction effect of treatment t on animal i in week j , and $animal_i \times week_j$ is the residual effect of animal i in week j . Lactation number ($Lact_i$) was used as a covariate for all variables.

In addition, in the case of DMI, milk yield and milk composition, BW and calculated energy balance the model also included as additional fixed effects (covariate) the deviation (for each of these traits) of each cow within each of the treatments 150, 160, and 170 g CP/kg DM from the mean of the wk 26 data for that treatment (i.e., the week before treatment groups being split). Cow was fitted as the subject factor and week was fitted as the time factor, with lactation number fitted as an additional fixed effect, with the model as follows:

$$y_{ij} = m + lact_i + cv_i + trt_t + week_j + trt \times week_{ij} + animal_i \times week_j,$$

where y_{ij} is the response to variable y for animal i in week j , m is the overall mean, $lact_i$ is the lactation effect for animal i , cv_i represents any number of covariate effects for animal i (number of covariates varies depending on response variable), trt_t is the main effect of treatment t on animal, $week_j$ is the main effect of week j , $trt \times week_{ij}$ is the interaction effect of treatment t on animal i in week j , and $animal_i \times week_j$ is the residual effect of animal i in week j . The correlation between time points was accounted for by an autoregressive model of order 1.

Milk FA concentration was analyzed using a linear model (REML estimation method) with the model as follows:

$$y_i = m + lact_j + trt_t + e_i,$$

where y_i is the response to variable y for animal i , m is the overall mean, $lact_j$ is the lactation effect for animal i , trt_t is the main effect of treatment t on animal i , and e_i is the residual effect of animal i .

One cow (170 g CP/kg DM treatment) did not complete the mid-late lactation nutrient utilization study due to health reasons (mastitis) and was subsequently treated as missing values during the statistical analysis. Mean intake, digestibility, N intake and output, as well as N utilization data were analyzed using a linear mixed model (REML estimation method) as follows:

$$y_{ij} = m + lact_j + trt_t + study_i + e_{ij},$$

where y_{ij} is the response to variable y for animal j in study i , m is the overall mean, $lact_j$ is the lactation effect for animal j (in study i), trt_t is the main effect of treatment t (150, 160, and 170 g CP/kg DM in early lactation and 140, 150, 160, and 170 g CP/kg DM in late lactation) on animal j (in study i), $study_i$ is the random effect of study i , and e_{ij} is the residual effect of animal j in study i .

In all analyses, if any of the weeks, treatments, or their interactions were significant ($P < 0.05$), then Fisher's

least significant difference test was used to compare pairwise the levels of individual fixed effects. The adequacy of all models was assessed by visual inspection of the appropriate residual plots.

RESULTS

Feed Composition

The mean chemical composition of feedstuffs offered are detailed in Table 2. The grass silage (mean ODM content of 357 g/kg, CP content of 134 g/kg DM, and ME of 11.6 MJ/kg DM) was typical of a good quality grass silage offered in Northern Ireland. The actual CP content of experimental diets (mean of 140, 152, 161, and 170 g CP/kg DM) were close to the formulated values of 140, 150, 160, and 170 g CP/kg DM, respectively. Concentrations of amino acids increased with increasing CP content of the experimental concentrates, as expected (Table 2).

Early to Mid Lactation (8–180 DIM)

MP Requirements. Based on actual cow performance, MP requirements were calculated to be 2,412, 2,501, and 2,487 g/d for cows offered diets containing 150, 160, and 170 g CP/kg DM. Therefore, diets containing 150, 160, and 170 g CP/kg DM were calculated to have supplied 99%, 98%, and 103%, respectively, of cows' MP intake when energy is limiting in the diet (**MP-E**) requirements, and 101%, 103%, and 111% of MP intake when nitrogen is limiting in the diet (**MP-N**) requirements (Table 3).

Intakes and Milk Production. Dietary CP content did not affect total DMI ($P = 0.082$), milk yield ($P = 0.536$), milk fat ($P = 0.746$), protein ($P = 0.240$) or lactose ($P = 0.812$) content, or yields of fat ($P = 0.258$), protein ($P = 0.116$), or fat plus protein ($P = 0.108$; Table 4). However, MUN concentration increased from 9.7 to 13.4 mg/dL with increasing diet CP content ($P < 0.001$), with significant differences between each treatment ($P < 0.05$). All production parameters varied over time ($P \leq 0.001$), but no treatment $\times$ week interactions were observed for any of the performance variables measured ($P > 0.05$), with the exception of milk protein content ($P < 0.001$, Figure 1). Lowering dietary CP content from 170 to 150 g CP/kg DM did not affect milk FA concentrations, except for the ratio of n-6 to n-3 FA, which was lower in milk from cows offered diets containing 150 and 160 g CP/kg DM compared with cows offered the diet containing 170 g CP/kg DM ($P < 0.001$), Table 4.

Body Tissue Reserves and Blood Metabolites. Reducing dietary CP content to either 150 or 160 g CP/kg DM decreased BCS ($P = 0.021$), and thickness of the longissimus dorsi loin (**LDL**) muscle ($P = 0.041$) compared with cows offered the 170 g CP/kg DM treatment, Table 5.

Dietary CP content did not affect cow BW ($P = 0.201$), backfat thickness ($P > 0.05$), and muscle thickness of GM ($P = 0.847$) and = longissimus dorsi thoracic (**LDT**; $P = 0.146$). Cow BW, BCS, backfat thickness at the LDL and muscle thickness at the GM and LDT varied over time ($P < 0.05$), but we found no treatment $\times$ week interactions for any of the body tissue variables measured. Calculated energy balance was greater for cows offered 150 compared with 160 g CP/kg DM treatment ($P < 0.001$) but neither differed from the 170 g/kg DM treatment. Energy balance varied over time ($P < 0.001$), but we did not observe a significant treatment $\times$ week interaction ($P = 0.830$).

Lowering dietary CP content from 170 to 150 g CP/kg DM reduced plasma albumin and plasma urea concentrations ($P = 0.013$ and $P < 0.001$, respectively), with significant differences between each treatment ($P < 0.05$; Table 5). Plasma globulin and total protein content was higher with the 170 treatment than with either of the 150 to 160 g CP/kg DM treatments ($P = 0.045$ and $P = 0.020$, respectively). All blood metabolites examined varied over time ($P < 0.001$), but there were no treatment $\times$ week interactions for any blood metabolites ($P > 0.05$). Total N intake increased with increasing total dietary CP content ($P < 0.001$), whereas milk N output was unaffected ($P = 0.110$). Nitrogen use efficiency was greater with the 150 and 160 g CP/kg DM treatment than with the 170 treatment ($P < 0.05$) (Table 5). Although N intake, milk N output, and NUE varied over time ($P < 0.001$), we found no treatment $\times$ week interactions ($P = 0.623$, $P = 0.556$, and $P = 0.659$, respectively).

Digestibility and Nitrogen Use Efficiency. For the subgroup of cows used in the nutrient utilization study, diet CP content had no effect on total DMI ($P = 0.411$), milk yield ($P = 0.375$), any of the digestibility coefficients examined ($P > 0.05$), total N intake ($P = 0.448$), N output in feces ($P = 0.099$), or milk N output ($P = 0.643$; Table 6). Lowering dietary CP content from 170 to 150 g CP/kg DM, decreased urinary N output ($P < 0.001$), with significant differences between each treatment ($P < 0.05$). Manure N output was reduced when dietary CP content was lowered from 170 or 160 to 150 g CP/kg DM ($P = 0.002$). Fecal N/N intake, manure N/N intake, and milk N/N intake were not affected by dietary CP content in early lactation ($P > 0.05$). Cows offered 170 compared with 150 g CP/kg DM had greater urinary N/N intake ($P = 0.017$) and urinary N/manure N ($P = 0.012$), but lower fecal N/manure N ($P = 0.012$).

Mid-Late Lactation (181–280 DIM)

MP Requirements. Based on actual cow performance, the MP requirements of cows offered diets containing 150, 160 and 170 g CP/kg DM were calculated to

Table 2. Chemical composition and amino acid composition of the grass silage, straw, and concentrates offered over the full experimental period

Item ²	Grass silage		Straw	Concentrate ¹									
	SD		SD	140	SD	150	SD	160	SD	170	SD	Parlor ³	SD
Oven DM (g/kg)	357	57.2	867	39.3	889	5.9	886	6.1	886	6.8	886	890	6.2
VCODM (g/kg)	367	56.6	—	—	—	—	—	—	—	—	—	—	—
CP (g/kg DM)	134	15.8	30	9.4	143	5.3	164	11.0	184	14.8	203	182	15.2
Ash (g/kg DM)	81	2.7	47	6.4	71	6.1	78	8.8	78	7.6	78	76	1.2
ADF (g/kg DM)	259	19.3	500	16.1	123	17.7	122	24.6	131	20.5	134	128	14.4
NDF (g/kg DM)	471	23.1	830	13.1	222	28.4	226	40.9	244	27.5	260	263	20.0
Starch (g/kg DM)	—	—	—	—	374	35.9	337	43.3	286	31.2	235	291	23.1
Gross energy (MJ/kg DM)	18.6	0.67	18.6	0.20	18.0	0.15	18.0	0.21	18.2	0.18	18.4	18.4	0.09
ME ⁴ (MJ/kg DM)	11.6	1.27	—	—	13.2	0	13.2	0	13.2	0.1	13.2	13.2	0
pH	4.0	0.26	—	—	—	—	—	—	—	—	—	—	—
Lactic acid (g/kg DM)	90.8	35.3	—	—	—	—	—	—	—	—	—	—	—
Acetic acid (g/kg DM)	14.3	5.88	—	—	—	—	—	—	—	—	—	—	—
Ethanol (g/kg DM)	10.1	4.42	—	—	—	—	—	—	—	—	—	—	—
Ammonia (g/kg total N)	62	11.7	—	—	—	—	—	—	—	—	—	—	—
EAA (g/100 g DM)													
Arg	0.29	0.04	0.06	0.02	0.78	0.04	0.93	0.06	1.03	0.08	1.14	0.92	0.10
His	0.20	0.01	0.03	0.02	0.39	0.01	0.45	0.02	0.49	0.03	0.54	0.46	0.07
Ile	0.52	0.02	0.06	0.02	0.55	0.02	0.64	0.03	0.70	0.05	0.79	0.57	0.18
Leu	0.88	0.03	0.12	0.03	1.16	0.03	1.35	0.05	1.48	0.10	1.63	1.44	0.10
Lys	0.53	0.03	0.08	0.03	0.74	0.05	0.83	0.06	0.94	0.06	1.02	0.82	0.11
Met	0.17	0.01	0.03	0.01	0.20	0.01	0.24	0.02	0.27	0.02	0.30	0.26	0.04
Phe	0.51	0.01	0.08	0.02	0.66	0.02	0.76	0.03	0.83	0.05	0.94	0.79	0.06
Thr	0.50	0.04	0.08	0.03	0.55	0.01	0.66	0.03	0.74	0.06	0.81	0.67	0.06
Val	0.70	0.04	0.09	0.02	0.71	0.01	0.81	0.03	0.89	0.04	1.00	0.83	0.03
NEAA (g/100 g DM)													
Cys	0.09	0.01	0.03	0.02	0.26	0.01	0.29	0.02	0.33	0.03	0.36	0.32	0.03
Ala	0.97	0.13	0.11	0.03	0.71	0.01	0.83	0.03	0.91	0.05	1.01	0.94	0.05
Asp	1.03	0.04	0.14	0.02	1.21	0.04	1.42	0.06	1.59	0.09	1.74	1.45	0.17
Glu	0.98	0.08	0.18	0.02	2.57	0.12	2.94	0.16	3.26	0.24	3.52	3.22	0.24
Gly	0.59	0.03	0.09	0.02	0.67	0.02	2.94	0.03	3.26	0.05	3.52	0.81	0.07
Ser	0.46	0.05	0.09	0.02	0.70	0.02	0.82	0.03	0.91	0.06	0.98	0.85	0.08
Tyr	0.33	0.01	0.04	0.01	0.52	0.03	0.57	0.02	0.63	0.04	0.70	0.59	0.04

¹Concentrates used to achieve total dietary CP concentrations of 140, 150, 160, and 170 g/kg DM.²VCODM = volatile-corrected oven DM.³Offered at 0.5 kg/cow per day to all cows in the parlor.⁴Metabolizable energy content of grass silage was predicted using infrared reflectance spectroscopy, and ME content of concentrates was calculated using DietCheck ration formulation software (DietCheck Advanced v10.1, DietCheck Ltd., Lancaster, UK).

Table 3. Calculated nutrient content of rations offered and MP requirements based on actual performance during the experiment, and calculated MP intakes when nitrogen (MP-N) and energy (MP-E) are limiting in the diet, during the early (8–180 DIM) and mid-late lactation (181–280 DIM) periods

Item	Total diet CP ¹ (g/kg DM)					
	150	150/140	160	160/140	170	170/140
8–180 DIM						
CP (g/kg DM)	151		161		172	
RDP (g)	2,177		2,200		2,257	
DUP (g)	1,022		1,154		1,288	
NDF (g/kg DM)	321		321		321	
ADF (g/kg DM)	229		229		228	
Starch (g/kg DM)	182		164		147	
ME (MJ/kg/DM)	12.3		12.3		12.3	
MP requirement ² (g/d)	2,412		2,501		2,487	
MP-N intake (g/d)	2,434		2,576		2,765	
MP-N (intake as % of requirements)	101		103		111	
MP-E intake (g/d)	2,395		2,462		2,574	
MP-E (intake as % of requirements)	99		98		103	
181–280 DIM						
CP (g/kg DM)	153	143	165	143	176	143
RDP (g)	2,189	1,951	2,276	1,922	2,388	1,879
DUP ³ (g)	969	846	1,144	826	1,317	797
NDF (g/kg DM)	326	326	327	326	328	326
ADF (g/kg DM)	227	227	227	227	227	227
Starch (g/kg DM)	185	201	165	201	146	201
ME (MJ/kg/DM)	12.3	12.3	12.3	12.3	12.3	12.3
MP requirement (g/d)	2,261	2,166	2,390	2,150	2,394	2,061
MP-N intake (g/d)	2,457	2,173	2,643	2,132	2,842	2,072
MP-N (intake as % of requirements)	109	100	111	99	119	101
MP-E intake (g/d)	2,363	2,169	2,462	2,128	2,580	2,062
MP-E (intake as % of requirements)	105	100	103	99	108	100

¹Total diet CP content in early/mid-late lactation.²Metabolizable protein requirement and supply as calculated for each experimental diet using DietCheck ration formulation software (DietCheck Advanced v10.1, DietCheck Ltd., Lancaster, UK) based upon actual animal performance.³DUP = digestible undegraded protein.

be 2,261, 2,390, and 2,394 g/d, respectively. Therefore, diets containing 150, 160, and 170 g CP/kg DM supplied 105%, 103%, and 108%, respectively, of cows' MP-E requirements, and 109%, 111%, and 119% of cows' MP-N requirements. The MP requirements of cows offered 150/140, 160/140, and 170/140 were calculated to be 2,166, 2,150, and 2,061 g/d, respectively, and these diets were calculated to have supplied 100%, 99%, and 100% of MP-E requirements and 100%, 99%, and 101% of MP-N requirements (Table 3).

Intake and Milk Production. Silage DMI and total DMI (Table 7) of cows that remained on the diet containing 150 g CP/kg DM was lower than for those that remained on the diet containing 170 g CP/kg DM in later lactation ($P = 0.05$). Reducing dietary CP content to 140 g CP/kg DM in mid-late lactation (150/140, 160/140, and 170/140) reduced total DMI compared with that of cows which remained on the 150, 160, or 170 g CP/kg DM treatments ($P < 0.001$). Cows on 170/140 treatment had a lower total DMI than cows on the 150/140 and 160/140 treatments ($P < 0.05$). During the mid-late lactation period, cows that remained on the diet containing 150 g

CP/kg DM produced less milk than those that remained on the diet containing 170 g CP/kg DM. Although mid-late lactation milk yield did not differ between treatments 150/140, 160/140, and 170/140 ($P > 0.05$), cows

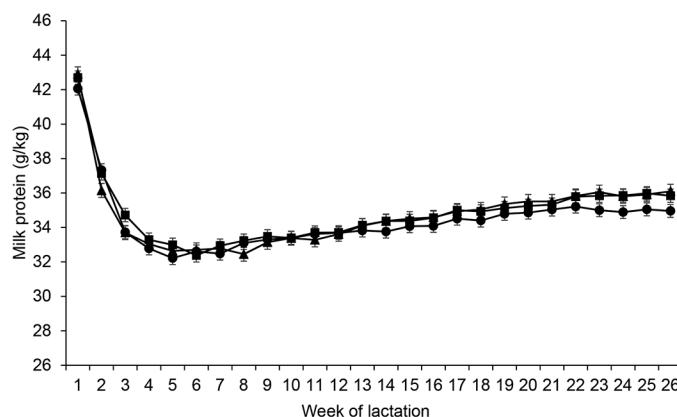
**Figure 1.** Mean weekly milk protein content (g/kg) of cows offered diets containing 150 (●), 160 (▲), or 170 (■) g CP/kg DM during 8 to 180 d of lactation. Error bars represent SEM.

Table 4. Effect of total dietary CP content from 8 to 180 DIM on DMI, milk production, milk composition, and on the fatty acid content of milk produced¹

Item	Total diet CP (g/kg DM)				P-value ²		
	150	160	170	SED	Trt	Week	Trt × week
DMI (kg/d)							
Silage	11.5	11.5	11.8	0.14	0.052	<0.001	0.905
Concentrate ³	11.5	11.5	11.8	0.15	0.169	<0.001	0.892
Straw	0.2	0.2	0.2	0.00	0.201	<0.001	0.374
Total	23.2	23.3	23.8	0.29	0.082	<0.001	0.862
Milk production							
Milk yield (kg/d)	35.7	37.1	36.3	0.97	0.536	<0.001	0.995
Fat (g/kg)	44.9	44.6	44.7	0.47	0.746	<0.001	0.593
Protein (g/kg)	34.4	34.8	34.9	0.28	0.240	<0.001	<0.001
Lactose (g/kg)	48.6	48.5	48.5	0.13	0.812	<0.001	0.602
Fat yield (kg/d)	1.60	1.64	1.62	0.027	0.258	<0.001	0.427
Protein yield (kg/d)	1.22	1.28	1.27	0.029	0.116	0.001	0.556
Fat + protein yield (kg/d)	2.82	2.92	2.89	0.051	0.108	<0.001	0.479
MUN (mg/dL)	9.7 ^a	11.5 ^b	13.4 ^c	0.18	<0.001	<0.001	0.094
Milk FA concentration (mg of fatty acid/mL of milk)							
C16:0	13.00	12.84	12.92	0.563	0.957	—	—
C18:0	3.46	3.52	3.79	0.176	0.151	—	—
C18:1 <i>cis</i> -9	6.09	6.35	6.68	0.237	0.056	—	—
CLA 18:2 <i>cis</i> -9, <i>trans</i> -11	0.15	0.16	0.16	0.007	0.083	—	—
C18:2 <i>cis</i> -9,12	0.67	0.67	0.71	0.023	0.151	—	—
C20:0	0.05	0.05	0.05	0.002	0.316	—	—
C18:3 <i>cis</i> -9,12,15	0.18	0.18	0.18	0.007	0.833	—	—
Total saturated	27.15	26.91	27.47	1.065	0.871	—	—
Total MUFA	7.22	7.51	7.87	0.283	0.084	—	—
Total PUFA	1.02	1.03	1.08	0.034	0.191	—	—
Total n-3 UFA	0.24	0.24	0.24	0.009	0.934	—	—
Total n-6 UFA	0.78	0.79	0.84	0.026	0.058	—	—
Total n-7 UFA	0.73	0.76	0.77	0.040	0.636	—	—
Total n-9 UFA	6.12	6.38	6.70	0.238	0.056	—	—
n-6/n-3 ratio	3.31 ^a	3.44 ^a	3.63 ^b	0.078	<0.001	—	—

^{a-c}Means within a row with no common superscripts differ ($P < 0.05$).

¹FA = fatty acids; SED = standard error of the difference.

²P-values: Trt = main effect of treatment; Week = main effect of week; Trt × week = interaction of treatment and week.

³Concentrate intake comprised the experimental concentrates offered in the PMR plus the parlor pellet offered at a fixed rate of 0.5 kg/cow per day.

on treatments 160/140 and 170/140 produced less milk than cows which remained on the 160 and 170 g CP/kg DM treatments in mid-late lactation. Milk fat content was unaffected by treatment from 181 to 280 DIM ($P = 0.267$). Milk protein content was greater for cows which remained on the 160 g CP/kg DM treatment in mid-late lactation compared with all other treatments ($P < 0.05$). In mid-late lactation, yields of fat, protein, and fat plus protein were lower with cows offered the 150 compared with the 160 or 170 g CP/kg DM treatment ($P < 0.001$), but did not differ between the 150/140, 160/140, and 170/140 treatments ($P > 0.05$). Cows offered the 150/140 and 170/140 treatments had lower fat, protein and fat plus protein yield than cows which remained on the diets containing 160 and 170 g CP/kg DM ($P < 0.05$). Milk urea nitrogen concentrations increased with each increment in total diet CP level (150, 160, and 170) and decreased across the 150/140, 160/140, and 170/140

treatments ($P < 0.05$). Dry matter intake, milk yield, and MUN all varied over time ($P < 0.05$, Figure 2a, 2b, and 2c, respectively). Milk fat, milk protein content, milk fat yield, and fat plus protein yield varied over time ($P < 0.05$).

Dietary CP content did not affect milk FA concentrations or ratios ($P > 0.05$) with the exception of CLA and n-6/n-3 ratio (Table 7). Proportions of CLA decreased when dietary CP level was reduced from 170 to 140 g CP/kg DM in mid-late lactation ($P = 0.003$). Cows on the 150 g CP/kg DM treatment had a lower n-6/n-3 FA ratio than those on the 160 and 170 g CP/kg DM treatment in mid-late lactation ($P < 0.05$), whereas this ratio did not differ between the 150/140, 160/140, and 170/140 treatments.

Body Tissue Reserves and Blood Metabolites. Dietary CP content did not affect BW ($P = 0.204$), backfat thickness ($P > 0.05$) or muscle thickness measured at the

Table 5. Effect of total dietary CP content from 8 to 180 DIM on body tissue reserves, energy balance, blood metabolites, and nitrogen use efficiency of dairy cows¹

Item	Total diet CP (g/kg DM)			SED	P-value ²		
	150	160	170		Trt	Week	Trt × week
BW (kg)	632	636	651	10.2	0.201	<0.001	0.929
BCS	2.4 ^a	2.4 ^a	2.5 ^b	0.04	0.021	0.003	0.705
Backfat thickness (mm)							
GM	6.4	6.2	6.2	0.22	0.162	0.221	0.223
LDL	6.1	5.9	6.3	0.14	0.080	<0.001	0.979
LDT	5.3	5.2	5.3	0.13	0.810	0.892	0.566
Muscle thickness (mm)							
GM	33.4	34.2	34.0	0.65	0.847	<0.001	0.393
LDL	33.0 ^a	33.0 ^a	33.5 ^b	0.31	0.041	0.225	0.574
LDT	40.1	40.1	41.3	0.77	0.146	<0.001	0.946
Energy balance ³ (MJ/d)	11.0 ^b	-4.2 ^a	6.2 ^{ab}	2.40	<0.001	<0.001	0.830
Blood plasma							
BHB (mmol/L)	0.34	0.35	0.35	0.02	0.562	<0.001	0.824
Glucose (mmol/L)	3.48	3.53	3.52	0.04	0.251	<0.001	0.519
NEFA (mmol/L)	0.26	0.25	0.25	0.02	0.446	<0.001	0.568
Total protein (g/L)	50.2 ^b	50.8 ^a	53.7 ^b	1.46	0.020	<0.001	0.773
Albumin (g/L)	28.1 ^a	28.9 ^b	30.2 ^c	0.75	0.013	<0.001	0.578
Globulin (g/L)	22.1 ^a	21.8 ^a	23.6 ^b	0.89	0.045	<0.001	0.962
Creatinine (μmol/L)	62.7	61.7	64.9	1.84	0.203	<0.001	0.680
Urea (mmol/L)	1.93 ^a	2.42 ^b	2.97 ^c	0.08	<0.001	<0.001	0.160
Nitrogen use efficiency							
Nitrogen intake (g/d)	568 ^a	602 ^b	648 ^c	7.18	<0.001	<0.001	0.623
Milk N output (g/d)	194	203	202	4.67	0.110	0.001	0.556
NUE ⁴	0.34 ^b	0.34 ^b	0.31 ^a	0.004	<0.001	<0.001	0.659

^{a-c}Means within a row with no common superscripts differ ($P < 0.05$).

¹GM = gluteus medius; LDL = longissimus dorsi loin; LDT = longissimus dorsi thoracic; NEFA = nonesterified fatty acids.

²P-values: Trt = main effect of treatment; Week = main effect of week; Trt × week = interaction of treatment and week.

³Energy balance calculated as energy balance (MJ of ME/d) = ME total intake – (ME maintenance + ME milk + ME pregnancy + ME activity).

⁴NUE = nitrogen use efficiency, calculated as milk N/intake N.

LDL ($P = 0.367$) or LDT ($P = 0.961$, Table 8). Cows that remained on the 170 g CP/kg DM treatment had greater BCS ($P = 0.005$) than those that remained on the 150 or 160 g CP/kg DM treatments. The BCS of cows on the 150/140, 160/140, and 170/140 treatments did not differ ($P > 0.05$), whereas the BCS of cows on each of these treatments did not differ from that of cows that remained on the original protein levels ($P > 0.05$). Muscle thickness of the GM was greater for cows offered 170 g CP/kg DM compared with cows offered the 150, 160, 160/140, and 170/140 treatments ($P = 0.018$), but it did not differ from that of cows that remained on 160 g CP/kg DM treatment ($P > 0.05$). Cows offered 170/140 treatment had lower energy balance compared with cows offered the 170, 150, and 150/140 treatment ($P < 0.001$), but did not differ from the 160 or 160/140 treatments. Cow BW, BCS, and energy balance varied over time ($P < 0.05$), but no treatment × week interactions were observed for any body tissue variables measured ($P > 0.05$).

Dietary CP content did not affect plasma albumin ($P = 0.060$), BHB ($P = 0.841$), NEFA ($P = 0.251$), or plasma glucose concentrations ($P = 0.426$, Table 8). Plasma cre-

atinine content did not differ between the 150, 160, and 170 g CP/kg DM treatments ($P > 0.05$), whereas cows on the 160/140 and 170/140 treatments had higher plasma creatinine concentrations than those on the 160 and 170 g CP/kg DM treatments ($P = 0.033$). Plasma globulin and total protein concentrations followed similar trends ($P = 0.019$ and $P = 0.022$, respectively), with concentrations tending to be greatest with the 160 and 170 g CP/kg DM treatments, whereas, with the exception of 160/140 treatment, concentrations did not fall when cows were moved to the 140 g CP/kg DM treatments. Plasma urea concentration increased with increasing diet CP levels (from 150 to 170 g CP/kg DM), and reducing dietary CP to 140 g CP/kg DM in mid-late lactation reduced plasma urea concentration ($P < 0.001$). Plasma urea concentrations did not differ between the 150/140, 160/140, and 170/140 treatments ($P > 0.05$). Blood metabolites varied over time ($P < 0.05$), but we found no treatment × week interactions for any of the blood metabolites measured ($P > 0.05$).

Total N intake (Table 8) of cows offered the 150/140, 160/140, and 170/140 treatments was lower than for

Table 6. Effect of total dietary CP content on nitrogen intake, nitrogen output, and nitrogen utilization efficiency of dairy cows during early lactation (mean 92 DIM)

Item	Total diet CP (g/kg DM)			SED	P-value
	150	160	170		
Silage DMI (kg/d)	11.7	11.7	10.9	0.64	0.408
Concentrate DMI (kg/d)	10.1	10.1	9.5	0.53	0.430
Total DMI (kg/d)	22.4	22.4	21.0	1.17	0.411
Milk yield (kg/d)	33.9	36.3	35.4	1.72	0.375
N intake and output (g/d)					
Total N intake	563	600	598	32.03	0.448
Fecal N	222	251	223	15.33	0.099
Urine N	138 ^a	161 ^b	185 ^c	9.84	<0.001
Manure N	360 ^a	412 ^b	408 ^b	16.40	0.002
Milk N	168	176	172	9.10	0.643
N utilization (g/g)					
Fecal N/N intake	0.398	0.420	0.365	0.0242	0.113
Urine N/N intake	0.249 ^a	0.270 ^{ab}	0.308 ^b	0.2111	0.017
Manure N/N intake	0.646	0.689	0.673	0.0296	0.341
Milk N/N intake	0.302	0.294	0.282	0.0123	0.279
Fecal N/manure N	0.616 ^b	0.605 ^b	0.546 ^a	0.0254	0.012
Urine N/manure N	0.384 ^a	0.395 ^a	0.454 ^b	0.0254	0.012
Digestibility coefficient (g/g)					
DM	0.763	0.763	0.772	0.0077	0.438
OM	0.781	0.783	0.792	0.0071	0.303
Nitrogen	0.602	0.581	0.635	0.0242	0.113
Gross energy	0.759	0.763	0.774	0.0082	0.189
ADF	0.734	0.754	0.757	0.0224	0.534
NDF	0.690	0.724	0.723	0.0229	0.273

^{a-c}Means within a row with no common superscripts differ ($P < 0.05$).

cows which remained on 150, 160, and 170 g CP/kg DM treatments ($P < 0.001$). Cows offered 170/140 and 160/140 treatments had significantly lower milk N output compared with cows which remained on diets containing 170 or 160 g CP/kg DM ($P < 0.001$), whereas milk N output did not differ between the 150/140 or 150 g CP/kg DM treatments ($P > 0.05$). Nitrogen use efficiency in mid-late lactation was lower with the 170 g CP/kg DM treatment than for either of the 150 or 160 g CP/kg DM treatments, whereas NUE was greatest with cows offered diets containing 140 g CP/kg DM (150/140, 160/140, 170/140). Total N intake, milk N output, and NUE all varied over time ($P < 0.05$), but we found no treatment $\times$ week interactions for these parameters ($P = 0.283$, $P = 0.517$, and $P = 0.524$, respectively).

Digestibility and Nitrogen Use Efficiency. For the subgroup of cows used in the nutrient utilization study, diet CP content did not affect total DMI ($P = 0.960$), milk yield ($P = 0.179$) or any of the digestibility coefficients ($P > 0.05$), with the exception of N digestibility ($P = 0.044$, Table 9). The latter was reduced for cows offered 140 g CP/kg DM compared with cows who were offered 150 or 160 g CP/kg DM ($P < 0.05$). Total N intake did not differ between treatments ($P = 0.284$). Cows offered 170 g CP/kg DM had a greater fecal N output ($P = 0.003$) compared with cows offered 140 and 150 g CP/kg DM. Manure N output decreased when dietary CP content was reduced from 170 to 140 g CP/kg DM ($P < 0.001$). Uri-

nary N output decreased with each incremental decrease in dietary CP content from 170 to 140 g CP/kg DM ($P < 0.001$). Dietary CP of 140 g CP/kg DM decreased milk N output compared with 160 or 170 g CP/kg DM ($P < 0.05$). Dietary CP content did not affect manure N/N intake ($P = 0.099$) or milk N/N intake ($P = 0.158$). Fecal N/N intake was greatest for cows offered 140 g CP/kg DM compared with cows offered 150 and 160 g CP/kg DM ($P = 0.044$), but did not differ from 170 g CP/kg DM treatment ($P > 0.05$). Cows offered 140 g CP/kg DM had greater fecal N/manure N ($P = 0.005$) but reduced urine N/N intake and urine N/manure N ($P < 0.001$ and $P = 0.005$, respectively), compared with cows offered 170 and 160 g CP/kg DM.

DISCUSSION

Relatively few studies have examined the impact of diet CP level over a full lactation. In one of the exceptions, which was undertaken at the Agri-Food and Biosciences Institute, Law et al. (2009) compared grass silage-based diets containing 173, 144, and 114 g CP/kg DM and observed a linear reduction in milk yield, milk fat yield, and milk fat plus protein yield with decreasing CP levels. Law et al. (2009) also demonstrated that total diet CP could be reduced to 144 g CP/kg DM in later lactation with no loss in performance. The current study was designed to examine the impact of offering diets

Table 7. Effect of total dietary CP content from 181 to 280 DIM on DMI, milk production, milk composition, and fatty acid content of milk produced¹

Item	Total diet CP ² (g/kg DM)						SED	P-value ³		
	150	150/140	160	160/140	170	170/140		Trt	Week	Trt × week
DMI (kg/d)										
Silage	11.6 ^{bc}	10.7 ^b	11.2 ^{cd}	10.6 ^b	11.6 ^d	10.1 ^a	0.24	<0.001	<0.001	0.431
Concentrate ⁴	11.7 ^c	11.1 ^b	11.8 ^c	11.0 ^b	12.0 ^c	10.5 ^a	0.22	<0.001	<0.001	0.579
Straw	0.2 ^{bc}	0.2 ^{bc}	0.2 ^{cd}	0.2 ^{ab}	0.2 ^d	0.2 ^a	0.01	<0.001	<0.001	0.090
Total	22.9 ^c	21.9 ^b	23.2 ^{cd}	21.8 ^b	23.9 ^d	20.9 ^a	0.45	<0.001	<0.001	0.651
Milk production										
Milk yield (kg/d)	30.5 ^{ab}	30.3 ^{ab}	32.2 ^{bc}	29.1 ^a	32.7 ^c	28.7 ^a	0.98	<0.001	<0.001	0.350
Fat (g/kg)	47.1	45.8	47.5	46.5	48.6	47.6	0.98	0.267	0.002	0.789
Protein (g/kg)	36.2 ^a	35.5 ^a	36.9 ^b	36.2 ^a	36.1 ^a	35.9 ^a	0.36	0.011	<0.001	0.856
Lactose (g/kg)	48.5 ^{bc}	48.6 ^{bc}	48.3 ^{ab}	48.2 ^a	48.3 ^{ab}	48.8 ^c	0.19	0.007	0.951	0.707
Fat yield (kg/d)	1.42 ^b	1.38 ^{ab}	1.55 ^c	1.35 ^a	1.56 ^c	1.36 ^{ab}	0.03	<0.001	<0.001	0.794
Protein yield (kg/d)	1.09 ^b	1.07 ^{ab}	1.21 ^c	1.05 ^{ab}	1.19 ^c	1.03 ^a	0.03	<0.001	<0.001	0.516
Fat + protein yield (kg/d)	2.51 ^a	2.45 ^a	2.76 ^b	2.40 ^a	2.74 ^b	2.41 ^a	0.06	<0.001	<0.001	0.646
MUN (mg/dL)	11.9 ^d	10.3 ^c	13.1 ^c	9.7 ^b	13.8 ^f	8.8 ^a	0.28	<0.001	0.016	0.625
Milk FA concentration (mg of fatty acid/mL of milk)										
C16:0	13.28	13.29	13.40	13.66	13.02	14.17	0.822	0.780	—	—
C18:0	3.85	3.76	3.64	3.90	4.17	3.71	0.295	0.564	—	—
C18:1 <i>cis</i> -9	7.16	6.89	7.10	7.25	7.67	6.94	0.335	0.220	—	—
CLA 18:2 <i>cis</i> -9, <i>trans</i> -11	0.16 ^{ab}	0.15 ^a	0.17 ^{bc}	0.15 ^{ab}	0.19 ^c	0.14 ^a	0.012	0.003	—	—
C18:2 <i>cis</i> -9,12	0.74	0.75	0.74	0.75	0.82	0.74	0.046	0.468	—	—
C20:0	0.06	0.05	0.05	0.06	0.06	0.05	0.004	0.539	—	—
C18:3 <i>cis</i> -9,12,15	0.20	0.21	0.18	0.21	0.21	0.21	0.012	0.218	—	—
Total SFA	28.03	27.93	28.19	28.59	28.22	28.44	1.583	0.998	—	—
Total MUFA	8.44	8.12	8.40	8.55	9.06	8.28	0.401	0.271	—	—
Total PUFA	1.15	1.16	1.13	1.17	1.25	1.15	0.063	0.535	—	—
Total n-3 UFA	0.27	0.27	0.24	0.28	0.27	0.27	0.014	0.259	—	—
Total n-6 UFA	0.88	0.89	0.89	0.90	0.97	0.88	0.051	0.402	—	—
Total n-7 UFA	0.85	0.78	0.84	0.88	0.92	0.88	0.061	0.268	—	—
Total n-9 UFA	7.18	6.91	7.12	7.27	7.70	6.96	0.337	0.220	—	—
n-6/n-3 ratio	3.2 ^a	3.3 ^a	3.7 ^b	3.3 ^a	3.6 ^b	3.3 ^a	0.123	<0.001	—	—

^{a-f}Means within a row with no common superscripts differ ($P < 0.05$).

¹FA = fatty acids.

²Total diet CP content in early/mid-late lactation.

³P-values: Trt = main effect of treatment; Week = main effect of week; Trt × week = interaction of treatment and week.

⁴Concentrate intake comprises the experimental concentrates offered in the PMR plus the parlor pellet offered at a fixed rate of 0.5 kg/cow per day.

containing 3 levels of diet CP in early lactation, when all diets met MP requirements. Similar to Law et al. (2009), the current study also examined the impact of reducing total diet CP level after mid lactation with a diet designed to meet MP requirements.

Performance During 8 to 180 DIM

Letelier et al. (2022) noted that cow performance responses associated with changes in diet CP content are sometimes due to changes in concentrate ingredients that are required to achieve the target CP levels, rather than changes in CP content per se. In the present study, we sought to maintain similar NDF (range, 226–260 g/kg DM) and ME contents (13.2 MJ/kg DM) within the concentrate component of the diet. However, concentrate starch levels decreased from 337 to 235 g/kg DM between the 150 and 170 treatment concentrates. Nevertheless, all treatments met the cows' MP-N requirements in

early lactation (101%, 103%, and 111% of requirements, based on actual performance, with 150, 160, and 170 g CP/kg DM, respectively), whereas the 150 and 160 g CP/kg DM diets marginally undersupplied MP-E (99%, 98%, 103% of requirements for 150, 160, and 170 g CP/kg DM, respectively, based on actual performance).

Although DMI is generally reduced when early lactation dairy cows are offered diets containing very low CP levels (Law et al., 2009; 114 g CP/kg DM), the impact of offering diets containing between 140 and 160 g CP/kg DM is less consistent. For example, total DMI was unaffected in early lactation studies when dietary CP content was reduced to 152 (Kalscheur et al., 1999), 154 (Wu and Satter, 2000), 144 (Law et al., 2009), 140 (Räsänen, 2016), and 153 g CP/kg DM (Zang et al., 2021), in agreement with the findings of the current study. In contrast, reduced intakes have been observed in studies involving diets containing 151 g CP/kg DM (Broderick, 2003) and both 140 and 160 g CP/kg DM (Reynolds et al., 2018),

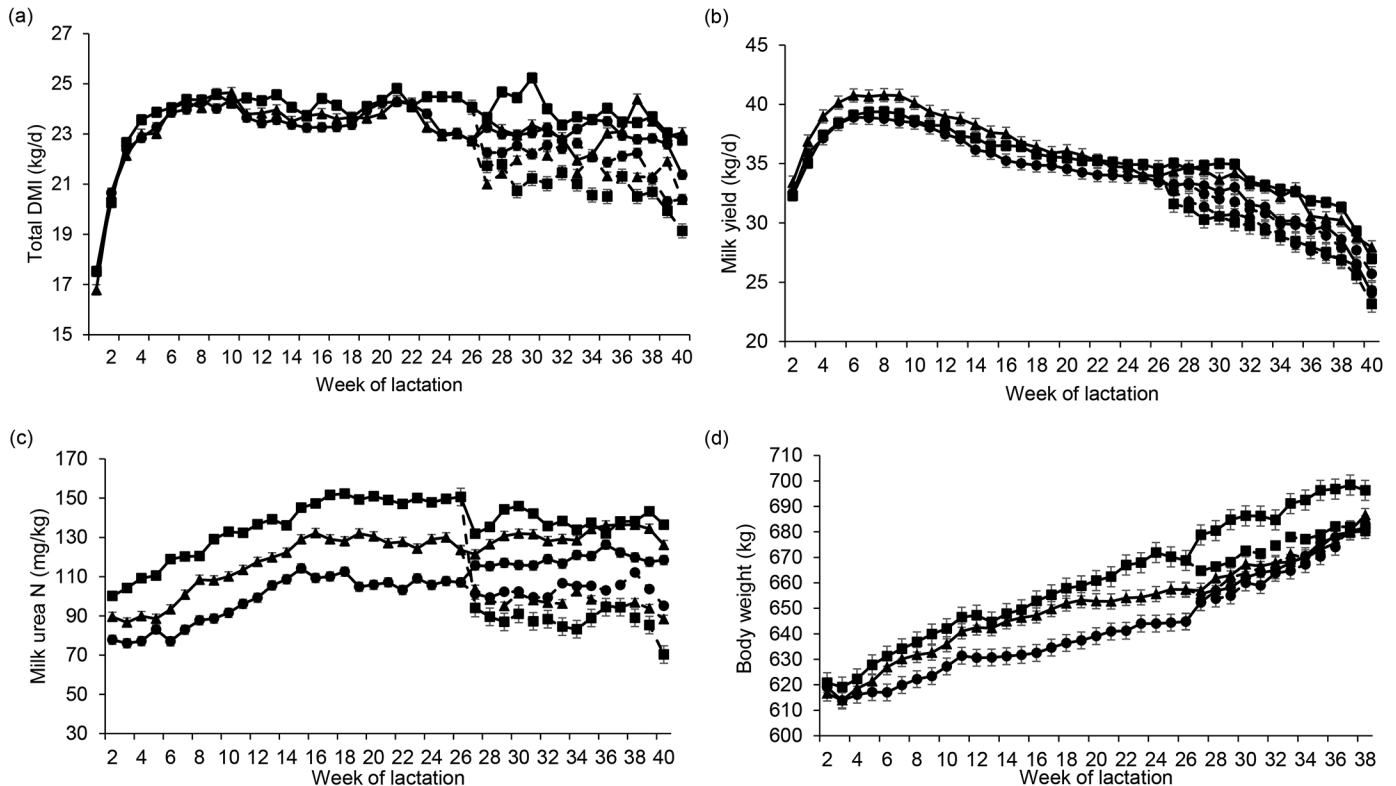


Figure 2. Mean weekly total DMI (kg/d; a), milk yield (kg/d; b), MUN (mg/kg; c), and BW (kg; d) of cows offered diets differing in total dietary CP content during 8 to 280 DIM, with solid lines representing dietary treatments offered for the full lactation period (150 [●], 160 [▲], and 170 [■] g/kg DM), and dashed lines representing cows that changed to the diet containing 140 g CP/kg DM at d 181 of lactation. Error bars represent SEM.

compared with diets containing >170 g CP/kg DM. It has been suggested that feeding diets low in RDP may restrict DMI due to limited rumen microbial growth, and restricted rumen digestion, in particular fiber digestion (Olmos Colmenero and Broderick, 2006; Lee et al., 2012). In the current study, the lowest protein diet (150 g CP/kg DM, with RDP averaging 9.2%) may have been marginal in RDP, although the minimum RDP level necessary to sustain rumen microbial function is still uncertain. It is also true that changes in CP levels are often achieved by modifying concentrate ingredients, which might affect nutrient digestibility and DMI (Nousiainen et al., 2009), although that was not the case in the current study. Although intakes with the 150 g CP/kg DM treatment tended to be lower than for the highest protein diet ($P = 0.082$), they did not differ from those of cows offered the 160 g CP/kg DM diet. Thus, the tendency for a higher intake with cows offered the diet containing 170 g CP/kg DM might reflect a response to the oversupply of MP with that treatment. Reynolds et al. (2018) also observed a greater DMI with a diet containing 180 g CP/kg DM which oversupplied MP requirements compared with a diet supplying 160 g CP/kg DM which met MP requirements). Indeed, Tebbe and Weiss (2021) observed

that offering a very high protein diet (200 g/kg DM) during the first 3 wk of lactation (with increased MP supply from a blend of RUP soy and rumen protected AA sources) had a positive effect on DMI (and ECM), which continues even after total diet CP was reduced to 170 g/kg DM. This study highlights that the balance or profile of AA in MP supply after calving may have longer-term implications.

A curvilinear milk yield response to increasing dietary CP content has been documented. For example, in a meta-analysis milk yield increased by 0.75 kg/d when dietary CP increased from 150 to 160 g CP/kg DM and by 0.35 kg/d when CP increased from 190 to 200 g CP/kg DM (NRC, 2001). Similarly, Ipharraguerre and Clark (2005) observed milk yield responses of 0.94 and 0.42 kg/d across the same diet CP levels. However, no milk yield response was observed in the current study, in agreement with Leonardi et al. (2003) and Olmos Colmenero and Broderick (2006) when dietary CP content was increased to >165 g CP/kg DM. The fact that milk yield was unaffected in the current study reflects, in part, the absence of an effect of diet CP level on DMI. Increased recycling of urea N from the blood stream to the gastro-intestinal tract provides an additional supply of N to rumen microbes,

Table 8. Effect of total dietary CP content from 181 to 280 DIM on body tissue reserves, energy balance, blood metabolites, and nitrogen use efficiency of dairy cows measured during mid-late lactation¹

Item	Total diet CP ² (g/kg DM)						SED	P-value ³		
	150	150/140	160	160/140	170	170/140		Trt	Week	Trt × week
BW (kg)	667	666	667	668	687	672	9.0	0.204	<0.001	0.807
BCS	2.5 ^a	2.4 ^a	2.5 ^a	2.5 ^a	2.7 ^b	2.6 ^{ab}	0.04	0.005	<0.001	0.160
Backfat thickness (mm)										
GM	6.46	5.99	6.01	6.72	6.20	6.26	0.430	0.441	0.131	0.778
LDL	5.74	6.16	6.06	6.34	7.00	6.40	0.393	0.061	0.266	0.905
LDT	5.41	5.08	5.55	5.37	5.54	5.40	0.586	0.647	0.267	0.489
Muscle thickness (mm)										
GM	31.5 ^{ab}	32.4 ^{bc}	31.7 ^{ab}	31.1 ^a	32.9 ^c	31.6 ^{ab}	0.561	0.018	0.411	0.490
LDL	33.02	32.09	33.60	32.30	33.46	32.86	0.835	0.367	0.783	0.104
LDT	38.63	38.09	38.16	37.99	37.96	39.49	2.032	0.961	0.951	0.269
Energy balance ⁴ (MJ/d)	28.5 ^c	20.2 ^{bc}	11.5 ^{ab}	15.7 ^{ab}	20.3 ^{bc}	7.9 ^a	4.52	<0.001	0.024	0.249
Blood plasma										
BHB (mmol/L)	0.44	0.47	0.48	0.45	0.46	0.45	0.032	0.841	<0.001	0.875
Glucose (mmol/L)	3.32	3.21	3.28	3.25	3.32	3.25	0.089	0.426	0.008	0.786
NEFA (mmol/L)	0.09	0.07	0.08	0.07	0.08	0.09	0.011	0.251	<0.001	0.210
Total protein (g/L)	60.7 ^a	61.7 ^{ab}	66.4 ^c	60.8 ^a	65.6 ^{bc}	61.5 ^{ab}	2.19	0.022	<0.001	0.791
Albumin (g/L)	32.9	34.4	35.6	34.0	35.5	34.2	1.04	0.060	<0.001	0.623
Globulin (g/L)	27.8 ^{ab}	27.0 ^{ab}	31.5 ^c	26.8 ^a	30.1 ^{bc}	27.3 ^{ab}	1.60	0.019	<0.001	0.947
Creatinine (μmol/L)	73.6 ^{ab}	75.7 ^{ab}	70.7 ^a	77.6 ^b	70.5 ^a	77.0 ^b	3.16	0.033	<0.001	0.563
Urea (mmol/L)	2.50 ^b	2.08 ^a	3.18 ^c	2.02 ^a	4.10 ^d	2.05 ^a	0.163	<0.001	<0.001	0.818
Nitrogen use efficiency										
Nitrogen intake (g/d)	565 ^c	496 ^b	612 ^d	490 ^{ab}	671 ^c	468 ^a	13.64	<0.001	0.001	0.283
Milk N output (g/d)	172 ^a	167 ^a	189 ^b	166 ^a	188 ^b	159 ^a	6.847	<0.001	<0.001	0.517
NUE ⁵	0.30 ^b	0.34 ^c	0.31 ^b	0.34 ^c	0.28 ^a	0.34 ^c	0.008	<0.001	0.009	0.524

^{a-c}Means within a row with no common superscripts differ ($P < 0.05$).

¹GM = gluteus medius; LDL = longissimus dorsi loin; LDT = longissimus dorsi thoracic; NEFA = nonesterified fatty acids.

²Total diet CP content in early/mid-late lactation.

³P-values: Trt = main effect of treatment; Week = main effect of week; Trt × week = interaction of treatment and week.

⁴Energy balance calculated as energy balance (MJ of ME/d) = ME total intake – (ME maintenance + ME milk + ME pregnancy + ME activity).

⁵NUE = nitrogen use efficiency, calculated as milk N/intake N.

and this can also help maintain milk production when diet protein levels are low (Mutsvangwa et al., 2016).

Although the impact of reducing diet CP content on milk protein content is not always consistent, an early review observed a 0.02% increase in milk protein for each 1% increase in diet protein (Jenkins and McGuire, 2006). In agreement with several longer-term studies (Law et al., 2009; Räisänen, 2016; Reynolds et al., 2018) milk protein content in the current study was unaffected by diet protein levels. This likely reflects the absence of an effect of diet on DMI and the relatively similar energy balance across treatments. Nevertheless, milk protein content of cows offered the lowest protein diet recovered at a slower rate from around wk 14 of lactation onwards than for cows offered diets containing either 160 or 170 g CP/kg DM. This might suggest that MP supply of cows on the low protein diet started to become limiting as the lactation progressed.

Although milk fat has been reported to both decrease (Law et al., 2009) and increase (Leonardi et al., 2003; Olmos Colmenero and Broderick, 2006) with increasing diet CP, milk fat content is generally unaffected by diet CP content, in agreement with the findings of the cur-

rent study. This likely reflects the similar fiber content of the concentrates offered. Although concentrate starch concentrations did differ across treatments, even with the 150 g CP/kg DM treatment total diet, starch was still relatively low (167 g/kg DM) and was unlikely to have affected milk fat. Given the absence of a treatment effect on milk yield, milk fat content or milk protein content, it was unsurprising that yields of fat, protein, or fat plus protein did not differ with increasing CP content of the diet.

When cows are offered grass silage-based diets, Nousiainen et al. (2004) suggested a MUN value of 11.7 mg/dL as being consistent with N requirements of rumen microbes being met, but Broderick et al. (2010) suggested 8.3 mg/dL as indicating ruminal N balance, with lower levels indicating RDP deficiency. The range in MUN concentrations (from 13.4 to 9.7 mg/dL) with decreasing dietary CP content in the current study align with target MUN concentrations (8–14 mg/dL) as defined by Kohn (2007) and Ishler (2016). The observed reduction of 1 to 2 mg/dL in MUN (per unit change in dietary CP) is similar to that described by Lavery and Ferris (2021). Therefore, even with the 150 g CP/kg DM diet, the MUN

Table 9. Effect of total dietary CP content on nitrogen intake, nitrogen output, and nitrogen utilization efficiency of dairy cows during mid-late lactation (mean 240 DIM)

Item	Total diet CP (g/kg DM)				SED	P-value
	140	150	160	170		
Silage DMI (kg/d)	10.2	10.6	10.7	10.2	1.14	0.960
Concentrate DMI (kg/d)	9.5	9.8	10.0	9.4	1.13	0.957
Total DMI (kg/d)	20.4	21.1	21.4	20.3	2.29	0.956
Milk yield (kg/d)	27.6	26.7	30.8	29.7	2.11	0.179
N intake and output (g/d)						
Total N intake	449	515	567	531	58.8	0.284
Fecal N	190 ^a	186 ^a	210 ^{ab}	242 ^b	15.9	0.003
Urine N	96 ^a	120 ^b	148 ^c	179 ^d	8.7	<0.001
Manure N	285 ^a	307 ^a	360 ^b	415 ^c	16.4	<0.001
Milk N	140 ^a	150 ^{ab}	170 ^b	170 ^b	12.2	0.019
N utilization (g/g)						
Fecal N/N intake	0.429 ^b	0.371 ^a	0.374 ^a	0.388 ^{ab}	0.0237	0.044
Urine N/N intake	0.214 ^a	0.239 ^{ab}	0.263 ^{bc}	0.295 ^c	0.0194	<0.001
Manure N/N intake	0.644	0.608	0.636	0.685	0.0274	0.099
Milk N/N intake	0.312	0.296	0.303	0.279	0.0147	0.158
Fecal N/manure N	0.665 ^b	0.608 ^{ab}	0.591 ^a	0.560 ^a	0.0303	0.005
Urine N/manure N	0.335 ^a	0.392 ^{ab}	0.410 ^b	0.440 ^b	0.0303	0.005
Digestibility coefficient (g/g)						
DM	0.772	0.786	0.782	0.784	0.0126	0.691
OM	0.788	0.801	0.799	0.803	0.0122	0.597
Nitrogen	0.572 ^a	0.629 ^b	0.626 ^b	0.612 ^{ab}	0.0237	0.044
Gross energy	0.764	0.781	0.777	0.780	0.0132	0.542
ADF	0.705	0.718	0.721	0.724	0.0171	0.691
NDF	0.675	0.692	0.690	0.711	0.0207	0.444

^{a-d}Means within a row with no common superscripts differ ($P < 0.05$).

concentration of 9.7 mg/dL suggests that RDP supply was adequate.

The effect of diet CP content on individual milk FA likely reflects the impact of individual concentrate ingredients rather than CP content per se (Cabrita et al., 2007). In the current study, greater concentrate CP levels were achieved by increasing the inclusion of rapeseed meal and protected soybean meal, with an associated decrease in starch and carbohydrate levels. These changes are likely responsible for the increase in C18:1 *cis*-9, total n-9, and n-6/n-3 ratio of FA observed. For example, trends for increased concentrations of C18:1 *cis*-9 (n-9 fatty acid) and C18:2 *cis*-9,12 (linoleic acid; n-6 fatty acid) were observed by Murphy et al. (1995) when cows were offered diets with supplemental rapeseed meal and soybean meal. Furthermore, the tendency for increased milk CLA (C18:2 *cis*-9,*trans*-11) content ($P = 0.083$), an intermediate product of rumen biohydrogenation, is likely due to increasing proportion of maize distillers as total diet CP content increased (Leonardi et al., 2005).

In general, diet had little effect on measures of body tissue, with only BCS and thickness of the LDL being moderately greater (+0.1 BCS and +0.5 mm) with the diet containing 170 g CP/kg DM. Although this may suggest that cows on this treatment may have mobilized less or deposited more body fat and muscle, there is no evidence of a combined impact of both, and the changes observed

here were very small, which supports the absence of observed differences in blood metabolites between treatments. The absence of an effect on BCS was not mirrored in a consistent effect on calculated energy balance, which was greater with the diet containing 150 g CP/kg DM than the diet containing 160 g CP/kg DM. Furthermore, the absence of an effect of diet in either plasma BHB or NEFA concentrations, both considered indicators of tissue mobilization (Cozzi et al., 2011; Benedet et al., 2019), and on plasma glucose concentrations, all suggest diet had no extreme effects on energy status. This is further supported by the absence of an effect of treatment on serum creatinine concentration, a marker for muscle and protein metabolism (Cozzi et al., 2011), with decreased creatinine concentrations in early lactation indicating skeletal muscle mobilization (Megahed et al., 2019; McCabe and Boerman, 2020). Blood total protein concentration, an indicator of intestinal protein absorption (Law et al., 2009), was higher with the higher protein diet, as expected. Plasma albumin levels can provide an indication of long-term protein status (Macrae et al., 2006). The findings of the current study agree with Raggio et al. (2007) and Law et al. (2009), who reported decreased plasma albumin levels with decreasing dietary CP content in early lactation, which Raggio et al. (2007) suggested may be due to decreased protein supply for hepatic protein synthesis. Plasma globulin levels, on the

other hand, can provide an indication of immune system function, so an absence of a treatment effect was not unexpected. The reduction in plasma urea levels with decreasing diet CP levels align with the effects of diet on milk urea levels and are also as expected (Law et al., 2009; Cabrita et al., 2011). Mean blood urea content for all treatments remained above 1.7 mmol/L, the concentration considered to be the lower limit under conditions in the United Kingdom (Macrae et al., 2006).

Reducing dietary CP content normally improves NUE of cows (Broderick, 2003; Hristov et al., 2004; Olmos Colmenero and Broderick, 2006), as at a lower dietary CP content, dietary N is more efficiently utilized for milk protein synthesis (Metcalf et al., 2008). In the current study NUE increased from 0.31 to 0.34 when dietary CP content was reduced from 170 to 150 g CP/kg DM, which is broadly in-line with improvements in NUE reported in other early lactation studies (Leonardi et al., 2003; Olmos Colmenero and Broderick, 2006; Law et al., 2009). However, contrary to the findings of others, no further improvement in NUE was observed when dietary CP was reduced below 160 g CP/kg DM, which may reflect the numerical lower milk protein yield with the diet containing 150 g CP/kg DM.

Nutrient Utilization in Early Lactation

The linear relationship between N intake and N excretion is well documented (Yan et al., 2006; Kebreab et al., 2010). In agreement with the findings of others (Castillo et al., 2000; Huhtanen et al., 2008; Weiss et al., 2009b; Kebreab et al., 2010), increasing dietary CP level increased urinary N excreted (g/d) and urinary N as a proportion of N intake, whereas fecal N output as a proportion of N intake was unaffected in the current study. The fact that manure N output did not increase between the 160 and 170 g CP/kg DM treatments reflects the numerically lower DMI with the latter in this subgroup of cows, which resulted in no further increase in N intake. When examined as a proportion of manure N output, the increase in fecal N and decrease in urine N excretion as diet CP level decreased is aligned with the findings of many other studies (Castillo et al., 2000; Huhtanen et al., 2008; Kebreab et al., 2010; Yang et al., 2022). This shift in N excretion with lower protein diets is environmentally advantageous, as ammonia release from fecal N is much lower than from urinary N (Dijkstra et al., 2013). Furthermore, these trends generally align with the NUE data, with urinary N being lower with the 150 and 160 g CP/kg DM treatments, which had the greatest NUE.

Digestibility of DM, organic matter, N, GE, NDF, and ADF have all been observed to decrease with decreasing dietary CP content and MP supply (Broderick et al., 2008; Weiss et al., 2009a; Lee et al., 2011; Räisänen et al.,

2022; Yang et al., 2022), with decreased OM digestibility having been attributed to a lower NDF digestibility. For example, in a meta-analysis by Huhtanen et al. (2009) and Nousiainen et al. (2009) NDF digestibility increased with dietary CP content, with an improvement in NDF digestibility of 0.77 g/kg per 1 g/kg of DM increase in diet CP content. It is possible that the absence of effects observed in the current study reflected the relatively small number of cows per treatment and insufficient statistical power to detect potentially very small differences. Regarding N digestibility, lowering dietary CP increases the proportion of endogenous protein excreted in feces, and this can contribute to decreased apparent digestibility of CP (Giallongo et al., 2015). In the current study, RDP supply was similar between treatments (decreasing from 9.7% to 9.2% between the 170 and 150 g CP/kg DM treatments), and although MP-N increased with increasing diet CP content, MP-E was marginal with the 2 lower protein treatments. Therefore, ruminal microbial activity and subsequent MP supply were likely similar across treatments. This may in part explain the absence of treatment effect on apparent digestibilities and supports the absence of a treatment effect on DMI and milk production.

Performance During 181 to 280 DIM

Although we observed a tendency for total DMI to be greater with the 170 g CP/kg DM diet in early lactation, the difference in intake between the 150 and 170 g CP/kg DM diets became significant in mid-late lactation. This might suggest that the 150 g CP/kg DM treatment, while perhaps just adequate in early lactation (meeting 99% MP requirements), was inadequate over a full lactation. However, although the ration was expected to supply 100% of MP requirements in mid-late lactation, based on actual performance, it actually oversupplied MP (105% of requirements). On this basis, it is possible that there was a positive DMI response to the oversupply of MP-N, with the 160 and 170 g CP/kg DM treatments supplying 111% and 119% of MP-N requirements, respectively. Supporting this, Reynolds et al. (2018) found that cows offered a diet containing 180 g CP/kg DM and which oversupplied MP had a consistently higher DMI than cows offered lower protein diets that met or were deficient in MP.

Similarly, although milk yield and milk constituent yields (fat, protein and fat plus protein) were unaffected by diet in early lactation, yields were reduced with the 150 CP treatment compared with the 170 g CP/kg DM treatment in mid-late lactation. This effect reflects a difference in milk yield, rather than differences in milk composition. The absence of an effect of diet on milk fat content in mid-late lactation is in agreement with the

findings of Barros et al. (2017) and consistent with early lactation findings. The high milk fat content observed across all treatments (47 g/kg) reflects the advanced DIM of cows at this point in the study. The greater milk protein content with the 160 g CP/kg DM treatment (compared with the 170 g CP/kg DM treatment), which occurred despite milk protein yield not being different, reflects the numerically greater milk yield with the 160 g CP/kg DM treatment. Although previous studies have shown no response to milk protein content with dietary CP content above 167 (Letelier et al., 2022), 162 (Barros et al., 2017), or 160 g CP/kg DM (Wu and Satter, 2000), in late lactation, milk protein content can be stimulated by specific AA supply, similar to the composition of casein or an increase in energy supply (Piccioli-Cappelli et al., 2014). The milk solids response to increasing diet CP is likely due to the concurrent increase in DMI.

In view of the findings of Law et al. (2009) that dietary CP content could be reduced from 173 to 144 g CP/kg DM in late lactation with no negative effects on DMI, milk yield or milk constituents, a similar response might have been expected in the current study. However, reducing diet CP from either 150, 160, or 170 g CP/kg DM to 140 g CP/kg DM reduced DMI in all treatments, with the reduction being greatest with the 170/140 treatment. There is no obvious explanation for the latter effect. Overall, the negative impact on performance with the 140 g CP/kg DM treatment may reflect the low RDP (8.9%) content of the diet which may have been inadequate to maintain rumen microbial function, and subsequent MP supply. The mid-late lactation nutrient utilization study provided evidence to support this, with N digestibility decreasing, and fiber digestibility showing a numerical (although nonsignificant) decrease with decreasing diet protein level.

Milk yield and fat plus protein yield in mid-late lactation did not differ between the 150 CP treatment and any of the 150/140, 160/140, or 170/140 CP treatments. Thus, for cows offered a low-protein diet in early lactation (150 g CP/kg DM), a further reduction in diet CP was possible with no negative effect on milk production. However, for cows offered higher CP diets in early lactation, milk production was reduced when these cows were offered lower protein diets in mid-late lactation. Similarly, Letelier et al. (2022) found that a dietary CP content of 167 g CP/kg DM maintained performance in later lactation, whereas cow performance was reduced when dietary CP was reduced to either 152 or 136 g CP/kg DM. Although Barros et al. (2017) found that decreasing dietary CP content in late lactation from 162 to 144 g CP/kg DM resulted in temporal decrease in milk yield and a marginal decline in milk protein yield, suggesting late lactation cows can adjust to a lower dietary CP content, there was no evidence of similar adjustment in the current study. This reduction

in performance occurred despite the diet containing 140 g CP/kg DM having been formulated to meet expected MP requirements and based on observed performance (lower DMI and production), having met 100% of MP requirements. This may reflect the limitations of current protein feeding systems, with Huhtanen and Ahvenjärvi (2022) suggesting that the majority of protein feeding systems currently overestimate MP supply from RUP, and the effects on rumen microbial protein synthesis are not fully elucidated. Therefore, although diets were formulated the supply 100% MP requirements, it is possible that MP supply was inadequate.

Furthermore, milk fat content, protein content, milk yield, and fat plus protein yield did not between the 150/140, 160/140, and 170/140 CP treatments, suggesting that there was no residual positive effect of offering higher protein diets in early lactation. In contrast, the greater the protein content of the diet offered in early lactation, the greater the reduction in MUN content with the 140 g CP/kg DM treatment in mid-late lactation. The reasons for this are unclear, but it is possible that cows offered a high-protein diet in early lactation had a reduced capacity for urea recycling when diet protein was reduced in mid-late lactation. Nevertheless, MUN concentrations during mid-late lactation (8.8 to 13.8 mg/dL) were still within the normal range of values defined by Kohn (2007), and although these concentrations suggest adequate protein supply, both Nousiainen et al. (2004) and Aguilar et al. (2012) have noted that few studies have examined MUN concentrations at low dietary CP content (<140 g CP/kg DM).

With the exception of CLA, diet had little effect on the FA profile of milk in late lactation. As dietary CP content increased, the content of CLA (C18:2, *cis*-9,*trans*-11), an intermediate product of biohydrogenation, increased, with this likely due to the increasing proportion of maize distillers in the diet (Leonardi et al., 2005). This effect was not apparent in the early to mid lactation and may reflect the changes in milk FA profile with advancing DIM (Kay et al., 2005). A numerically greater concentration of linoleic acid with increasing dietary CP content is likely responsible for the greater n-6/n-3 ratio of FA with increasing dietary CP.

The higher BCS of cows in the 170 g CP/kg DM treatment in early lactation continued into mid-late lactation, with cows on this treatment having a higher BCS and greater GM muscle thickness. However, none of the BW, BCS, backfat or muscle thickness data suggest that difference in performance between treatments could be attributed to difference in body tissue mobilization or deposition. This is supported by similar NEFA, BHB, and plasma glucose concentrations between treatments, and is in agreement with the findings of Law et al. (2009). Although apparent calculated energy balance data were

variable between treatments and not particularly reflective of other indicators of energy status, there was evidence that cows offered higher protein diets in early lactation exhibited a lesser energy balance when moved to a low protein diet in mid-late lactation (20.2, 15.7, and 7.9 MJ/d for 150/140, 160/140, and 170/140, respectively), with this reflecting the effects on DMI.

The greater plasma creatinine concentrations when cows changed from the 160 and 170 g CP/kg DM treatment in early lactation to the 140 g CP/kg DM treatment in mid-late lactation suggest skeletal muscle deposition, although it is not expected that this would have resulted in appreciable changes in body tissue reserves. As expected (Kauffman and St-Pierre, 2001; Kohn et al., 2005; Olmos Colmenero and Broderick, 2006), and in common with early lactation findings, blood urea levels continued to reflect diet CP content in mid-late lactation, decreasing from 4.1 to 2.5 mmol/L when dietary CP content was reduced from 170 to 150 g CP/kg DM. However, in contrast to MUN, protein level in early lactation had no effect on blood urea levels in mid-late lactation, with levels not differing between the 150/140, 160/140, or 170/140 treatments. Given the normally close relationship between MUN and blood urea, the reason for this difference is unclear. The lower or trend for lower serum globulin and total protein concentrations with the 150 and 140 CP treatments is indicative of lower intestinal absorption of protein (Law et al., 2009).

Mean NUE across the 150, 160, and 170 g CP/kg DM diets was numerically lower in mid-late lactation than in early lactation (proportionally 0.04, 0.03, and 0.03, respectively), perhaps reflecting the impacts of protein mobilization from body tissue in early lactation versus protein deposition in mid-late lactation. Furthermore, the oversupply of MP in mid-late lactation may have reduced the efficiency of MP utilization for milk true protein synthesis (Metcalf et al., 2008; Arriola Apelo et al., 2014; Daniel et al., 2016). The NUE of 0.34 with the diets containing 140 g CP/kg DM demonstrates the potential of low protein diets to improve NUE, albeit at the expense of cow performance.

Nutrient Utilization in Mid-Late Lactation

Reducing diet CP from 170 to 140 g CP/kg DM reduced N output in feces, urine, and manure by 21%, 46%, and 31%, respectively. The increase in fecal N/N intake and fecal N/manure N with lower protein diets partly reflects the lower digestibility of N as diet protein levels decreases, the latter in agreement with previous studies (Weiss et al., 2009a; Giallongo et al., 2015; Räisänen et al., 2022; Yang et al., 2022). In the current study the diet containing 140 g CP/kg DM was low in RDP content (8.8%) and therefore N supply to ruminal microbes may

have been limited and N digestibility impaired. The shift to lower urinary N excretion (urinary N/N intake and urinary N/manure N) with lower protein diets is frequently observed (Katongole and Yan, 2020; Yang et al., 2022), and is desirable from an environmental perspective (Waldrup et al., 2013).

CONCLUSIONS

Lowering dietary CP content from 170 to 150 g CP/kg DM while continuing to meet MP requirements had no detrimental effect on DMI, milk yield, milk composition, or yield of milk constituents in early lactation (8–180 DIM), and it reduced N excretion and improved NUE. In mid-late lactation (181–280 DIM), fat plus protein yield was lower for cows offered diets containing 150 g CP/kg DM compared with those offered diets containing either 160 or 170 g CP/kg DM. Decreasing dietary CP content to 140 g CP/kg DM in mid-late lactation had no negative effect on cow performance compared with the 150 g CP/kg DM treatment, but it improved NUE. The findings of this study suggest that even when MP requirements are fully met, a dietary CP content of 150 g CP/kg DM may be marginal over a full lactation, and that a dietary CP of 160 g/kg DM is desirable.

NOTES

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Nonstandard abbreviations used: AFBI = Agri-Food and Biosciences Institute; DUP = digestible undegraded

protein; FA = fatty acid; FAME = FA determined as methyl esters; FiM = Feed into Milk; GE = gross energy; GM = gluteus medius; LDL = longissimus dorsi loin; LDT = longissimus dorsi thoracic; MP-E = MP intake when energy is limiting in the diet; MP-N = MP intake when nitrogen is limiting in the diet; NEFA = nonesterified fatty acids; NUE = nitrogen use efficiency; ODM = oven DM; SED = standard error of the difference; Trt = treatment; VCODM = volatile-corrected oven DM.

REFERENCES

- Agnew, R., T. Yan, J. France, E. Kebreab, and C. Thomas. 2004. Energy requirement and supply. Pages 11–20 in *Feed into Milk: A New Applied Feeding System for Dairy Cows*. C. Thomas, ed. Nottingham University Press, Nottingham, UK.
- Aguilar, M., M. D. Hanigan, H. Tucker, B. Jones, S. Garbade, M. L. McGilliard, C. C. Stallings, K. F. Knowlton, and R. James. 2012. Cow and herd variation in milk urea nitrogen concentrations in lactating dairy cattle. *J. Dairy Sci.* 95:7261–7268. <https://doi.org/10.3168/jds.2012-5582>.
- Arriola Apelo, S. I., J. Knapp, and M. Hanigan. 2014. Invited review: Current representation and future trends of predicting amino acid utilization in the lactating dairy cow. *J. Dairy Sci.* 97:4000–4017. <https://doi.org/10.3168/jds.2013-7392>.
- Barros, T., M. Quaassdorff, M. Aguerre, J. O. Colmenero, S. Bertics, P. Crump, and M. Wattiaux. 2017. Effects of dietary crude protein concentration on late-lactation dairy cow performance and indicators of nitrogen utilization. *J. Dairy Sci.* 100:5434–5448. <https://doi.org/10.3168/jds.2016-11917>.
- Benedet, A., C. Manuelian, A. Zidi, M. Penasa, and M. De Marchi. 2019. Invited review: β -hydroxybutyrate concentration in blood and milk and its associations with cow performance. *Animal* 13:1676–1689. <https://doi.org/10.1017/S175173111900034X>.
- Bligh, E. G., and W. J. Dyer. 1959. A rapid method of total lipid extraction and purification. *Can. J. Biochem. Physiol.* 37:911–917. <https://doi.org/10.1139/o59-099>.
- Bougouin, A., A. Leytem, J. Dijkstra, R. S. Dungan, and E. Kebreab. 2016. Nutritional and environmental effects on ammonia emissions from dairy cattle housing: A meta-analysis. *J. Environ. Qual.* 45:1123–1132. <https://doi.org/10.2134/jeq2015.07.0389>.
- Broderick, G. A. 2003. Effects of varying dietary protein and energy levels on the production of lactating dairy cows. *J. Dairy Sci.* 86:1370–1381. [https://doi.org/10.3168/jds.S0022-0302\(03\)73721-7](https://doi.org/10.3168/jds.S0022-0302(03)73721-7).
- Broderick, G. A., P. Huhtanen, S. Ahvenjärvi, S. Reynal, and K. Shingfield. 2010. Quantifying ruminal nitrogen metabolism using the omasal sampling technique in cattle—A meta-analysis. *J. Dairy Sci.* 93:3216–3230. <https://doi.org/10.3168/jds.2009-2989>.
- Broderick, G. A., M. Stevenson, R. Patton, N. Lobos, and J. J. Olmos Colmenero. 2008. Effect of supplementing rumen-protected methionine on production and nitrogen excretion in lactating dairy cows. *J. Dairy Sci.* 91:1092–1102. <https://doi.org/10.3168/jds.2007-0769>.
- Cabrita, A. R., R. Bessa, S. Alves, R. Dewhurst, and A. Fonseca. 2007. Effects of dietary protein and starch on intake, milk production, and milk fatty acid profiles of dairy cows fed corn silage-based diets. *J. Dairy Sci.* 90:1429–1439. [https://doi.org/10.3168/jds.S0022-0302\(07\)71628-4](https://doi.org/10.3168/jds.S0022-0302(07)71628-4).
- Cabrita, A. R., R. Dewhurst, D. Melo, J. Moorby, and A. Fonseca. 2011. Effects of dietary protein concentration and balance of absorbable amino acids on productive responses of dairy cows fed corn silage-based diets. *J. Dairy Sci.* 94:4647–4656. <https://doi.org/10.3168/jds.2010-4097>.
- Castillo, A., E. Kebreab, D. Beever, and J. France. 2000. A review of efficiency of nitrogen utilisation in lactating dairy cows and its relationship with environmental pollution. *J. Anim. Feed Sci.* 9:1–32. <https://doi.org/10.22358/jafs/68025/2000>.
- Chen, X., G. Finney, H. Zheng, H. Wang, A. W. Gordon, C. P. Ferris, E. Magowan, and T. Yan. 2020. Modern Holstein-origin dairy cows within grassland-based systems partition more feed nitrogen into milk and excrete less in manure. *Sci. Total Environ.* 727:138561. <https://doi.org/10.1016/j.scitotenv.2020.138561>.
- Cozzi, G., L. Ravarotto, F. Gottardo, A. Stefani, B. Contiero, L. Moro, M. Brscic, and P. Dalvit. 2011. Reference values for blood parameters in Holstein dairy cows: Effects of parity, stage of lactation, and season of production. *J. Dairy Sci.* 94:3895–3901. <https://doi.org/10.3168/jds.2010-3687>.
- Daniel, J. B., N. C. Friggens, P. Chapoutot, H. Van Laar, and D. Sauvant. 2016. Milk yield and milk composition responses to change in predicted net energy and metabolizable protein: a meta-analysis. *Animal* 10:1975–1985. <https://doi.org/10.1017/S1751731116001245>.
- Dijkstra, J., O. Oenema, J. Van Groenigen, J. Spek, A. Van Vuuren, and A. Bannink. 2013. Diet effects on urine composition of cattle and N₂O emissions. *Animal* 7(Suppl. 2):292–302. <https://doi.org/10.1017/S1751731113000578>.
- Edmonson, A., I. Lean, L. Weaver, T. Farver, and G. Webster. 1989. A body condition scoring chart for Holstein dairy cows. *J. Dairy Sci.* 72:68–78. [https://doi.org/10.3168/jds.S0022-0302\(89\)79081-0](https://doi.org/10.3168/jds.S0022-0302(89)79081-0).
- Firkins, J., and C. Reynolds. 2005. Whole-animal nitrogen balance in cattle. Pages 167–186 in *Nitrogen and Phosphorus Nutrition of Cattle: Reducing the Environmental Impact of Cattle Operations*. E. Pfeffer and A. N. Hristov, ed. CABI Publishing, Wallingford, UK.
- Giallongo, F., A. N. Hristov, J. Oh, T. Frederick, H. Weeks, J. Werner, H. Lapiere, R. A. Patton, A. Gehman, and C. Parys. 2015. Effects of slow-release urea and rumen-protected methionine and histidine on performance of dairy cows. *J. Dairy Sci.* 98:3292–3308. <https://doi.org/10.3168/jds.2014-8791>.
- Hristov, A. N., R. Etter, J. Ropp, and K. Grande. 2004. Effect of dietary crude protein level and degradability on ruminal fermentation and nitrogen utilization in lactating dairy cows. *J. Anim. Sci.* 82:3219–3229. <https://doi.org/10.2527/2004.82113219x>.
- Hristov, A. N., M. Hanigan, A. Cole, R. Todd, T. McAllister, P. Ndegwa, and A. Rotz. 2011. Ammonia emissions from dairy farms and beef feedlots. *Can. J. Anim. Sci.* 91:1–35. <https://doi.org/10.4141/CJAS10034>.
- Huhtanen, P., and S. Ahvenjärvi. 2022. Problems in determining metabolisable protein value of dairy cow diets and the impact on protein feeding. *Animal* 16:100539. <https://doi.org/10.1016/j.animal.2022.100539>.
- Huhtanen, P., J. I. Nousiainen, M. Rinne, K. Kytölä, and H. Khalili. 2008. Utilization and partition of dietary nitrogen in dairy cows fed grass silage-based diets. *J. Dairy Sci.* 91:3589–3599. <https://doi.org/10.3168/jds.2008-1181>.
- Huhtanen, P., M. Rinne, and J. Nousiainen. 2009. A meta-analysis of feed digestion in dairy cows. 2. The effects of feeding level and diet composition on digestibility. *J. Dairy Sci.* 92:5031–5042. <https://doi.org/10.3168/jds.2008-1834>.
- Ipharraguerre, I. R., and J. H. Clark. 2005. Impacts of the source and amount of crude protein on the intestinal supply of nitrogen fractions and performance of dairy cows. *J. Dairy Sci.* 88:E22–E37. [https://doi.org/10.3168/jds.S0022-0302\(05\)73134-9](https://doi.org/10.3168/jds.S0022-0302(05)73134-9).
- Ishler, M. 2016. Interpretation of Milk Urea Nitrogen (MUN) Values. Penn State Extension. Accessed Nov. 2, 2020. <https://extension.psu.edu/interpretation-of-milk-urea-nitrogen-mun-values>.
- Jenkins, T. C., and M. A. McGuire. 2006. Major advances in nutrition: Impact on milk composition. *J. Dairy Sci.* 89:1302–1310. [https://doi.org/10.3168/jds.S0022-0302\(06\)72198-1](https://doi.org/10.3168/jds.S0022-0302(06)72198-1).
- Kalscheur, K. F., J. H. Vandersall, R. A. Erdman, R. A. Kohn, and E. Russek-Cohen. 1999. Effects of dietary crude protein concentration and degradability on milk production responses of early, mid, and late lactation dairy cows. *J. Dairy Sci.* 82:545–554. [https://doi.org/10.3168/jds.S0022-0302\(99\)75266-5](https://doi.org/10.3168/jds.S0022-0302(99)75266-5).
- Katongole, C. B., and T. Yan. 2020. Effect of varying dietary crude protein level on feed intake, nutrient digestibility, milk production, and nitrogen use efficiency by lactating Holstein-Friesian cows. *Animals (Basel)* 10:2439. <https://doi.org/10.3390/ani10122439>.
- Kauffman, A. J., and N. R. St-Pierre. 2001. The relationship of milk urea nitrogen to urine nitrogen excretion in Holstein and Jersey cows. *J. Dairy Sci.* 84:2284–2294. [https://doi.org/10.3168/jds.S0022-0302\(01\)74675-9](https://doi.org/10.3168/jds.S0022-0302(01)74675-9).

- Kay, J. K., W. J. Weber, C. E. Moore, D. E. Bauman, L. B. Hansen, H. Chester-Jones, B. A. Crooker, and L. H. Baumgard. 2005. Effects of week of lactation and genetic selection for milk yield on milk fatty acid composition in Holstein cows. *J. Dairy Sci.* 88:3886–3893. [https://doi.org/10.3168/jds.S0022-0302\(05\)73074-5](https://doi.org/10.3168/jds.S0022-0302(05)73074-5).
- Kebreab, E., A. Strathe, J. Dijkstra, J. A. Mills, C. K. Reynolds, L. A. Crompton, T. Yan, and J. France. 2010. Energy and protein interactions and their effect on nitrogen excretion in dairy cows. Pages 417–425 in *Proceedings of the 3rd EAAP International Symposium on Energy and Protein Metabolism*. Wageningen Academic Publishers, the Netherlands.
- Kohn, R. 2007. Use of milk or blood urea nitrogen to identify feed management inefficiencies and estimate nitrogen excretion by dairy cattle and other animals. Pages 30–31 in *Proc. Florida Ruminant Nutrition Symposium*. University of Florida, Gainesville, FL.
- Kohn, R. A., M. M. Dinneen, and E. Russek-Cohen. 2005. Using blood urea nitrogen to predict nitrogen excretion and efficiency of nitrogen utilization in cattle, sheep, goats, horses, pigs, and rats. *J. Anim. Sci.* 83:879–889. <https://doi.org/10.2527/2005.834879x>.
- Lapierre, H., and G. Lobley. 2001. Nitrogen recycling in the ruminant: A review. *J. Dairy Sci.* 84:E223–E236. [https://doi.org/10.3168/jds.S0022-0302\(01\)70222-6](https://doi.org/10.3168/jds.S0022-0302(01)70222-6).
- Lavery, A., and C. P. Ferris. 2021. Proxy measures and novel strategies for estimating nitrogen utilisation efficiency in dairy cattle. *Animals (Basel)* 11:343. <https://doi.org/10.3390/ani11020343>.
- Lavery, A., A. Gordon, A. White, and C. Ferris. 2023. Performance of dairy cows offered either zero-grazed grass or grass silage prepared from the same sward. *Animal* 17:101003. <https://doi.org/10.1016/j.animal.2023.101003>.
- Law, R. A., F. J. Young, D. C. Patterson, D. J. Kilpatrick, A. R. G. Wylie, and C. S. Mayne. 2009. Effect of dietary protein content on animal production and blood metabolites of dairy cows during lactation. *J. Dairy Sci.* 92:1001–1012. <https://doi.org/10.3168/jds.2008-1155>.
- Lee, C., A. Hristov, K. Heyler, T. Cassidy, M. Long, B. Corl, and S. Karnati. 2011. Effects of dietary protein concentration and coconut oil supplementation on nitrogen utilization and production in dairy cows. *J. Dairy Sci.* 94:5544–5557. <https://doi.org/10.3168/jds.2010-3889>.
- Lee, C., A. N. Hristov, T. Cassidy, K. Heyler, H. Lapierre, G. Varga, M. De Veth, R. Patton, and C. Parys. 2012. Rumen-protected lysine, methionine, and histidine increase milk protein yield in dairy cows fed a metabolizable protein-deficient diet. *J. Dairy Sci.* 95:6042–6056. <https://doi.org/10.3168/jds.2012-5581>.
- Leonardi, C., S. Bertics, and L. Armentano. 2005. Effect of increasing oil from distillers grains or corn oil on lactation performance. *J. Dairy Sci.* 88:2820–2827. [https://doi.org/10.3168/jds.S0022-0302\(05\)72962-3](https://doi.org/10.3168/jds.S0022-0302(05)72962-3).
- Leonardi, C., M. Stevenson, and L. Armentano. 2003. Effect of two levels of crude protein and methionine supplementation on performance of dairy cows. *J. Dairy Sci.* 86:4033–4042. [https://doi.org/10.3168/jds.S0022-0302\(03\)74014-4](https://doi.org/10.3168/jds.S0022-0302(03)74014-4).
- Letelier, P., G. Zanton, and M. Wattiaux. 2022. Production performance of Holstein cows at 4 stages of lactation fed 4 dietary crude protein concentrations. *J. Dairy Sci.* 105:9581–9596. <https://doi.org/10.3168/jds.2022-22146>.
- Liu, E., M. Hanigan, and M. VandeHaar. 2021. Importance of considering body weight change in response to dietary protein deficiency in lactating dairy cows. *J. Dairy Sci.* 104:11567–11579. <https://doi.org/10.3168/jds.2020-19566>.
- Macrae, A. I., D. A. Whitaker, E. Burrough, A. Dowell, and J. Kelly. 2006. Use of metabolic profiles for the assessment of dietary adequacy in UK dairy herds. *Vet. Rec.* 159:655–661. <https://doi.org/10.1136/vr.159.20.655>.
- McCabe, C. J., and J. P. Boerman. 2020. Invited Review: Quantifying protein mobilization in dairy cows during the transition period. *Appl. Anim. Sci.* 36:389–396. <https://doi.org/10.15232/aas.2019-01929>.
- Megahed, A. A., M. W. H. Hiew, D. Ragland, and P. D. Constable. 2019. Changes in skeletal muscle thickness and echogenicity and plasma creatinine concentration as indicators of protein and intramuscular fat mobilization in periparturient dairy cows. *J. Dairy Sci.* 102:5550–5565. <https://doi.org/10.3168/jds.2018-15063>.
- Metcalf, J. A., R. J. Mansbridge, J. S. Blake, J. D. Oldham, and J. R. Newbold. 2008. The efficiency of conversion of metabolisable protein into milk true protein over a range of metabolisable protein intakes. *Animal* 2:1193–1202. <https://doi.org/10.1017/S1751731108002140>.
- Murphy, J., J. Connolly, and G. McNeill. 1995. Effects on cow performance and milk fat composition of feeding full fat soyabeans and rapeseeds to dairy cows at pasture. *Livest. Prod. Sci.* 44:13–25. [https://doi.org/10.1016/0301-6226\(95\)00048-P](https://doi.org/10.1016/0301-6226(95)00048-P).
- Mutsaers, T., K. Davies, J. McKinnon, and D. Christensen. 2016. Effects of dietary crude protein and rumen-degradable protein concentrations on urea recycling, nitrogen balance, omasal nutrient flow, and milk production in dairy cows. *J. Dairy Sci.* 99:6298–6310. <https://doi.org/10.3168/jds.2016-10917>.
- Nousiainen, J., M. Rinne, and P. Huhtanen. 2009. A meta-analysis of feed digestion in dairy cows. 1. The effects of forage and concentrate factors on total diet digestibility. *J. Dairy Sci.* 92:5019–5030. <https://doi.org/10.3168/jds.2008-1833>.
- Nousiainen, J., K. Shingfield, and P. Huhtanen. 2004. Evaluation of milk urea nitrogen as a diagnostic of protein feeding. *J. Dairy Sci.* 87:386–398. [https://doi.org/10.3168/jds.S0022-0302\(04\)73178-1](https://doi.org/10.3168/jds.S0022-0302(04)73178-1).
- NRC. 2001. *Nutrient Requirements of Dairy Cattle*. 7th rev. ed. National Academies Press, Washington, DC.
- Olmos Colmenero, J. J., and G. Broderick. 2006. Effect of dietary crude protein concentration on milk production and nitrogen utilization in lactating dairy cows. *J. Dairy Sci.* 89:1704–1712. [https://doi.org/10.3168/jds.S0022-0302\(06\)72238-X](https://doi.org/10.3168/jds.S0022-0302(06)72238-X).
- Park, R., R. Agnew, F. Gordon, and R. Steen. 1998. The use of near infrared reflectance spectroscopy (NIRS) on undried samples of grass silage to predict chemical composition and digestibility parameters. *Anim. Feed Sci. Technol.* 72:155–167. [https://doi.org/10.1016/S0377-8401\(97\)00175-2](https://doi.org/10.1016/S0377-8401(97)00175-2).
- Piccioli-Cappelli, F., J. Looor, C. Seal, A. Minuti, and E. Trevisi. 2014. Effect of dietary starch level and high rumen-undegradable protein on endocrine-metabolic status, milk yield, and milk composition in dairy cows during early and late lactation. *J. Dairy Sci.* 97:7788–7803. <https://doi.org/10.3168/jds.2014-8336>.
- Raggio, G., G. Lobley, R. Berthiaume, D. Pellerin, G. Allard, P. Dubreuil, and H. Lapierre. 2007. Effect of protein supply on hepatic synthesis of plasma and constitutive proteins in lactating dairy cows. *J. Dairy Sci.* 90:352–359. [https://doi.org/10.3168/jds.S0022-0302\(07\)72636-X](https://doi.org/10.3168/jds.S0022-0302(07)72636-X).
- Räisänen, S. 2016. Milk production and protein utilization in high producing dairy cows fed a low CP diet based on cereals and roughage only: A full lactation study. Master's thesis. Department of Animal Nutrition and Management, Swedish University of Agricultural Sciences, Uppsala, Sweden.
- Räisänen, S. E., C. Lage, C. Zhou, A. Melgar, T. Silvestre, D. Wasson, S. Cueva, J. Werner, T. Takagi, M. Miura, and A. N. Hristov. 2022. Lactational performance and plasma and muscle amino acid concentrations in dairy cows fed diets supplying 2 levels of digestible histidine and metabolizable protein. *J. Dairy Sci.* 105:170–187. <https://doi.org/10.3168/jds.2021-20800>.
- Reynolds, C., A. Thomson, A. Jones, and L. Crompton. 2018. Protein for dairy cows: How low is achievable? Long term effects of reduced protein supply. *Feed Compd.* 38:38–41.
- Tebbe, A. W., and W. P. Weiss. 2021. Concurrent and carryover effects of feeding blends of protein and amino acids in high-protein diets with different concentrations of forage fiber to fresh cows. 1. Production and blood metabolites. *J. Dairy Sci.* 104:5583–5600. <https://doi.org/10.3168/jds.2020-19663>.
- Thomas, C. 2004. *Feed into Milk: A New Applied Feeding System for Dairy Cows*. Nottingham University Press, Nottingham, United Kingdom.
- Van Soest, P. 1976. Laboratory methods for evaluating the energy value of feedstuffs. Pages 83–94 in *Feed Energy Sources for Livestock*. H. Swan and D. Lewis, ed. Butterworths, London, United Kingdom.
- Waldrip, H. M., R. W. Todd, and N. A. Cole. 2013. Prediction of nitrogen excretion by beef cattle: A meta-analysis. *J. Anim. Sci.* 91:4290–4302. <https://doi.org/10.2527/jas.2012-5818>.
- Wattiaux, M., M. Uddin, P. Letelier, R. Jackson, and R. Larson. 2019. Invited Review: Emission and mitigation of greenhouse gases from

- dairy farms: The cow, the manure, and the field. *Appl. Anim. Sci.* 35:238–254. <https://doi.org/10.15232/aas.2018-01803>.
- Weiss, W. P., N. St-Pierre, and L. Willett. 2009a. Varying type of forage, concentration of metabolizable protein, and source of carbohydrate affects nutrient digestibility and production by dairy cows. *J. Dairy Sci.* 92:5595–5606. <https://doi.org/10.3168/jds.2009-2247>.
- Weiss, W. P., L. Willett, N. St-Pierre, D. Borger, T. McKelvey, and D. Wyatt. 2009b. Varying forage type, metabolizable protein concentration, and carbohydrate source affects manure excretion, manure ammonia, and nitrogen metabolism of dairy cows. *J. Dairy Sci.* 92:5607–5619. <https://doi.org/10.3168/jds.2009-2248>.
- Wu, Z., and L. Satter. 2000. Milk production during the complete lactation of dairy cows fed diets containing different amounts of protein. *J. Dairy Sci.* 83:1042–1051. [https://doi.org/10.3168/jds.S0022-0302\(00\)74968-X](https://doi.org/10.3168/jds.S0022-0302(00)74968-X).
- Yan, T., J. Frost, R. Agnew, R. Binnie, and C. Mayne. 2006. Relationships among manure nitrogen output and dietary and animal factors in lactating dairy cows. *J. Dairy Sci.* 89:3981–3991. [https://doi.org/10.3168/jds.S0022-0302\(06\)72441-9](https://doi.org/10.3168/jds.S0022-0302(06)72441-9).
- Yang, C., C. Ferris, and T. Yan. 2022. Effects of dietary crude protein concentration on animal performance and nitrogen utilisation efficiency at different stages of lactation in Holstein-Friesian dairy cows. *Animal* 16:100562. <https://doi.org/10.1016/j.animal.2022.100562>.
- Zang, Y., P. Ji, S. Y. Morrison, Y. Koba, R. J. Grant, and H. M. Dann. 2021. Reducing metabolizable protein supply: Effects on milk production, blood metabolites, and health in early-lactation dairy cows. *J. Dairy Sci.* 104:12443–12458. <https://doi.org/10.3168/jds.2021-20459>.

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